

EDNS 151 Final Report

Reformers

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1.0 Problem Validation, User Empathy, and Stakeholder Engagement

1.1 PROBLEM STATEMENT

How might we reduce the waste produced by single-use packaging materials?

1.2 BACKGROUND/CONTEXT

Plastic and styrofoam have been used in packaging since the 1800s to secure products in custom-made containers. Although practical, these materials are non-biodegradable and represent a major portion of landfill waste. Studies indicate that styrofoam can take up to 500 years to decompose, during which it leeches harmful chemicals into the surrounding environment [1]. In addition, global plastic waste management practices are also insufficient, with only 5% of all plastic waste being recycled as of 2019 and 86% of plastic waste ending up in landfills [2]. These materials have greatly impacted the environment, and there is a need for an innovative approach that reduces waste and supports sustainable packaging alternatives.

1.3 KEY STAKEHOLDERS

- Consumers: They are the end users of the packaging, and its design impacts their experience (e.g., ease of use, durability, and appeal)
- Manufacturers: They are responsible for producing the packaging, and its design directly affects their production processes, costs, and material sourcing
- Retailers: The packaging affects how easily products can be stored, transported, and displayed
- Environmental Advocates: These groups monitor and advocate for sustainable practices
- Regulatory Bodies: They ensure the packaging complies with legal standards, such as food safety, recyclability, or biodegradability

1.4 EXISTING SOLUTIONS

There are not only a number of different products but strategies towards dealing with the issue of waste in packaging. The two main thoughts are that of redesign and materials engineering. The latter is the thought that altering the materials to reduce waste by keeping the same packaging idea, the common making envelope for example, a tried and true classical design. In the form of recycled cardboard or plastics or other materials. This strategy is an intriguing one because one could still use pre-existing infrastructure but could still use sustainable materials. Some of the key faults of existing pieces are their strength, one only needs to look at paper straws as an example and relate them to packaging, partial but not full recyclability, and cost are the huge factors at play for faults in this design space. The other, a reinvention of packaging for sustainable design is a more novel idea and requires more research to reasonably tackle.

1.5 ANCILLARY ISSUES

Many outside issues arise when bringing up the topic of sustainable packaging. Some difficulties arise from how sustainable packaging is more expensive or how more sustainable packaging takes up more space. Another problem is that only 2% of plastic packaging is actually recycled and on top of that about “30% of plastic-based packaging items are either too complex or too small to recycle”[3]. Lastly there are plenty of government regulations, like the Fair Packaging and Labeling Act (FPLA) and the Uniform Packaging and Labeling Regulation (UPLR) [4]. As more and more uses for packaging increases these problems will arise alongside this increase.

1.6 GROUP CITATIONS

- [1] S. Coelho, “Why styrofoam is so hard to recycle and what you can do about it,” *Business Insider*. <https://www.businessinsider.com/guides/home/is-styrofoam-recyclable>
- [2] C. Cassella, “Turns Out All That Plastic Currently Sitting in US Landfills Is Worth BILLIONS of Dollars,” *ScienceAlert*, May 03, 2022. <https://www.sciencealert.com/all-that-plastic-that-ends-up-in-landfill-is-worth-billions-scientists-say>
- [3] I. D. Ibrahim et al., “Need for Sustainable Packaging: an Overview,” *Polymers*, vol. 14, no. 20, p. 4430, 2022, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9609329/>
- [4] “Packaging and Labeling,” NIST, Jan. 2010, Available: <https://www.nist.gov/pml/owm/packaging-and-labeling>
- [5] Team Logo, ChatGPT <https://chat.openai.com> Prompt: “Please come up with a team logo for my design team. Our team is one that is attempting to reduce packaging waste.”

1.7 INDIVIDUAL SECTIONS

1.7.1 - Chase Pellegrom

1.7.1.1 - Stakeholder Outreach Summary

Packaging plays a crucial role in both e-commerce and retail, influencing operations across various industries. To gain insight into daily packaging practices, an in-person survey was conducted at the Colorado Mills Mall in Lakewood, Colorado. The survey focused on three stores: VisionWorks, LEGO, and Crocs.

The first survey was conducted with VisionWorks managers, Ayesha and Brandon. When asked about their daily shipment volume, they estimated receiving approximately two to three large boxes per day, each containing 12 to 15 pairs of glasses. Each pair arrives in its standard protective case, which is then placed inside an individual box. These individual boxes are stacked within a larger box, lined with bubble wrap interwoven between layers for added protection. While discussing packaging challenges, Ayesha and Brandon emphasized the need to balance between minimizing materials and ensuring product safety. While they acknowledged the importance of sustainability, they were only aware of the company's recycling initiatives, primarily encouraging customers to recycle. However, further research revealed that VisionWorks does have a glasses recycling program, though no specific efforts to reduce packaging waste were found [1]. Regarding the functional and aesthetic aspects of their packaging, they noted that appearance is not a primary concern since glasses are sold in their protective cases rather than their individual boxes. When asked about potential changes, both managers expressed that the amount of packaging used could be reduced, specifically suggesting that the bubble wrap could be eliminated or minimized without compromising product safety.

Vision Works Shipment Standard

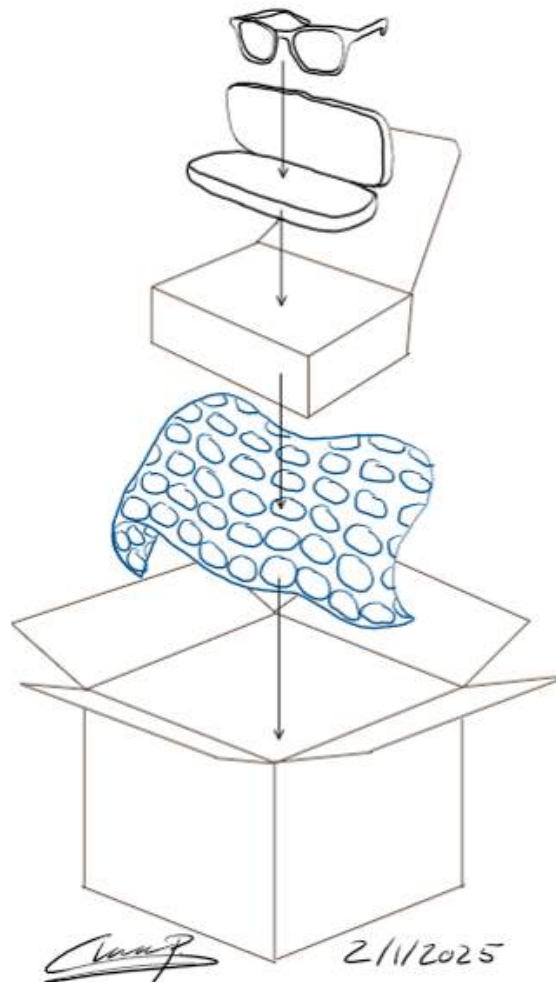


Figure 1: VisionWorks Packaging

The next in-person survey was conducted at LEGO, where manager Jackson Wisler provided insights into their packaging and sustainability efforts. According to Wisler, LEGO receives truck deliveries approximately four times a week, with each truck carrying five pallets of varying box quantities. When asked about packaging challenges, he noted that the primary issue stems from weather-related delays, particularly during the winter months when shipments from their Texas-based distribution center are sometimes postponed. However, there was no mention of challenges regarding the product packaging itself. When asked about sustainability efforts, Wisler emphasized that LEGO is highly committed to eco-friendly practices. The company aims to replace plastic bags inside its sets with paper by 2026, optimize box sizes to reduce material waste, and has strongly promoted recycling. Additionally, LEGO is investing in renewable

energy systems for its factories. Further research revealed that the company also has goals of achieving net-zero carbon emissions by 2050, transitioning to more sustainable product materials by 2032 (instead of the current resin they use), and is tripling its sustainability investments to \$1.4 billion in 2025 [2]. Regarding the balance between functionality and attractiveness, Wisler said that the outside of the product box needs to have an image, and the packets on the inside need to be easily opened, so they often have rip tabs. When asked about potential packaging improvements, he suggested that LEGO should accelerate the transition to paper bags within sets and could always reduce the amount of cardboard used in shipping.

The final in-person survey was conducted at Crocs with manager Leanne. When asked about their shipment volume, she explained that it varies based on previous sales and projected demand. On average, Crocs receives 2-3 shipments per week, with each shipment containing 3-5 boxes, and each box holding around five pairs of shoes. Regarding packaging challenges, Leanne noted a minor issue with space optimization. Since Crocs are lightweight and do not require extra protection, each pair is packaged in a simple plastic bag rather than a box. However, when these bags are placed in shipping boxes, there often seems to be excess space, leading to inefficiencies in packaging and transport. When asked about sustainability, Leanne mentioned that 25% of the material in each pair of Crocs comes from recycled plant-based sources, intending to increase this to 50% (by 2030). Further research revealed additional sustainability efforts, including their development of a polybag for shipping that has an 85% lower carbon footprint than traditional shoeboxes and a recycling program for old Crocs shoes [3]. When discussing the balance between functionality and attractiveness, she emphasized that this is not a major concern, as the packaging consists solely of a recyclable bag. As for potential improvements, Leanne suggested that better space optimization in shipping boxes could help reduce waste and improve efficiency.

To gain further insight into packaging challenges across different retail environments, an interview and in-person walkthrough were conducted at the Adidas branch store at the Outlets at Castle Rock with Manager Alexis Alvarez. Unlike the previously surveyed stores, which primarily sell a single type of product, Adidas carries a broader range of items, including shoes, clothing, and specialty products like soccer balls and water bottles. The store receives three standard shipments per week, but once a month, they receive a significantly larger shipment containing over 250 boxes of shoes, clothing, and specialty items. Packaging waste is a major issue across these product categories. Each shoebox contains a paper liner around the shoes and wadded paper inside for support. One notable challenge the store faces is that damaged shoeboxes on the sales floor must be replaced, although rare, this results in material waste and inconvenience for store managers.

Clothing items are individually wrapped in single-use plastic, which, although highly encouraged for recycling (as indicated on the packaging itself), contributes significantly to waste. The store explained that this plastic wrapping is necessary to maintain the pre-folded condition of the clothing, preventing wrinkles during shipping. However, once received, most of these garments are immediately removed from the plastic, unfolded, and either hung up or refolded to match store display designs. Additionally, many items, such as jackets and jerseys, contain extra paper linings, further adding to unnecessary waste.

Miscellaneous items, including water bottles, lunch boxes, hats, soccer balls, etc., are also individually wrapped in plastic, which is collected and recycled at the end of each day. One of the store's biggest packaging-related challenges is the volume of waste that accumulates. By the

end of the night, it can require a team of four people approximately 45 minutes to transfer all the plastic and cardboard waste to the mall's recycling bin.

Adidas is deeply committed to sustainability and has set many targets to reduce its environmental impact across various aspects of its supply chain. By 2025, the company aims for 90% of its products to be made from sustainable materials, with a strong focus on recycled and renewable resources. Adidas is working toward climate neutrality across its operations by 2025 and net-zero carbon emissions by 2050, implementing energy-efficient practices and increasing the use of renewable energy. Adidas has replaced virgin polyester with recycled alternatives in all products in 2024. The company is also focused on circularity, aiming to develop products designed for reuse, repair, and recyclability [4].

The in-person surveys at VisionWorks, LEGO, Crocs, and Adidas revealed a clear pattern of plastic overuse across various industries. VisionWorks relies on bubble wrap to protect fragile glasses during transit, LEGO uses plastic bags inside their product boxes to keep pieces organized, and Crocs packages their shoes in plastic poly bags. Adidas further contributes to plastic waste by individually wrapping clothing, accessories, and shoes, even though much of this packaging is immediately discarded upon arrival. This information has revealed a common theme: the primary reason for excessive packaging is product protection during shipping.

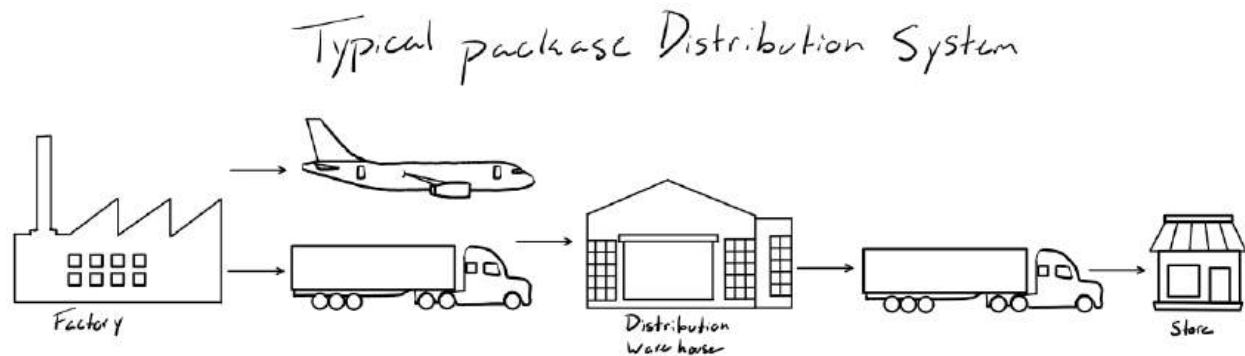


Figure 2: Package Delivery System

Companies prioritize minimizing damage, keeping items grouped together, and ensuring their products arrive in good condition. However, findings suggest that many of these packaging choices are not optimized for waste reduction. This confirms that while product protection is a necessary concern, the materials and methods used can be reassessed for sustainability.

This stakeholder outreach directly supports the problem statement: "How might we reduce the waste produced by single-use packaging materials?" The excessive reliance on single-use plastics and unnecessary packaging materials validates the design challenge and highlights clear areas for improvement. Companies acknowledge the need for sustainability but require solutions that maintain product safety, organization, and efficiency. Therefore, the team's design process will focus on developing alternative cushioning and protective materials that reduce the need for bubble wrap and plastic while still ensuring product safety during shipping. Additionally, the team will explore sustainable materials for internal packaging, for example, replacing LEGO's plastic bags with paper or biodegradable alternatives, and work on optimizing shipping box efficiency to reduce wasted space and excess materials, as seen with Crocs and Adidas.

These insights emphasize that single-use plastic has become the default option, not necessarily the best one, presenting an opportunity for better alternatives. Companies appear willing to adopt more sustainable practices, but they need packaging solutions that do not compromise product protection or organization. The design will prioritize both functionality and sustainability, ensuring that retailers can reduce waste without sacrificing efficiency or product integrity. Moving forward, the team will explore innovative, sustainable packaging solutions that address the balance between protection, organization, and waste reduction, ultimately contributing to a significant reduction in single-use packaging waste in retail and e-commerce.

1.7.1.2 - Remaining Unknowns

While this research has provided valuable insights into packaging waste across multiple retail stores, several key unknowns remain. There is still a need to understand the full extent of packaging decision-making at the corporate level, including how much flexibility individual stores have in modifying their packaging materials and whether sustainability initiatives influence real-world packaging choices. Additionally, while companies emphasize recycling, it is unclear how much of their packaging waste is actually recycled versus discarded and whether recycling infrastructure at the store level is effective in reducing overall waste. Another unknown is how customers perceive and interact with packaging waste (do they prefer minimal packaging, or do they expect certain protective materials or designs?) Understanding customer expectations could help shape solutions that balance waste reduction with consumer experience. Additionally, while it was identified that shipping protection is a major concern for retailers, this stakeholder outreach did not provide data on alternative protective materials that could be sustainable, protective, and cost-effective. Exploring viable materials that maintain product integrity while reducing waste is crucial to the design criteria. Lastly, it is not yet fully understood the logistical and financial constraints of implementing sustainable packaging alternatives (are companies open to adopting new materials if they meet cost and performance requirements, or are there strict budget limitations that prevent innovation?) These unknowns highlight the need for further investigation into corporate policies, consumer behavior, recycling effectiveness, and alternative materials.

1.7.1.3 - Summary

Stakeholder research has confirmed that packaging waste is a significant issue across various retail industries, with excessive use of single-use plastics and unnecessary materials. Through in-person surveys and interviews at VisionWorks, LEGO, Crocs, and Adidas, there was an observed pattern of plastic overuse primarily driven by the need for product protection during shipping. This stakeholder outreach validates the problem statement: "How might we reduce the waste produced by single-use packaging materials?" The excessive reliance on plastic and inefficient packaging methods confirms that the team's design challenge is a critical issue that needs attention. Companies acknowledge the importance of sustainability but require solutions that still provide product protection, organization, and efficiency. The team's design process will focus on developing alternative protective materials that eliminate the need for bubble wrap and plastic while maintaining product safety. Findings indicate that single-use plastic remains the industry default, not because it is the best option but because it is the most convenient. This presents an opportunity for innovative solutions that balance sustainability with functionality.

Companies appear willing to adopt more sustainable packaging if it meets their needs, making this a problem worth solving. Moving forward, the team will focus on developing sustainable packaging solutions that address protection, organization, and waste reduction, ultimately contributing to a significant decrease in single-use packaging waste across retail and e-commerce.

1.7.1.4 - Individual Bibliography

[1] VisionWorks, “Can eyewear be eco-friendly? - visionworks,” VisionWorks, <https://www.visionworks.com/article-eco-friendly-eyewear-options> (accessed Feb. 1, 2025).

[2] LEGO, “Sustainability/home - sustainability,” LEGO.com, <https://www.lego.com/en-us/sustainability> (accessed Feb. 1, 2025).

[3] Crocs, “Comfort for the Planet,” Crocs.com, <https://www.crocs.com/comfort-for-the-planet.html> (accessed Feb. 1, 2025).

[4] Adidas, “Our targets - Adidas group,” Adidas, <https://www.adidas-group.com/en/sustainability/focus-on-sustainability/our-targets> (accessed Feb. 1, 2025).

1.7.2 - Quentin Tischler

1.7.2.1 - Research Summary

Documentary analysis: “Unwrapping - the documentary” by Éco Entreprises Québec (EEQ)

This documentary by (EEQ) focuses on the impact packaging has in consumer’s lives from commercial to household effects. The core thought of the designers focusing on sustainable packing in the film is not to remove packaging all together but to redesign it in a sustainable manner. It is impossible to simply remove packaging from the world due to its necessity for preservation of perishables and other goods that need to be protected. Unfortunately this makes the design challenge increasingly difficult because you want something that won't parish and protect something, but will eventually parish to decompose. It's a rather hypocritical statement which makes it difficult for designers to tackle. While the average consumer commonly thinks about what they need, they don't often think about the packaging they are buying their product in. One that may seem very foreign to an American audience would be milk in non-recyclable plastic bags that are placed in pitchers which are common in Canada, presented in the film. This is one such example of stakeholder blindness to problems. CONsumers simply want quick and cheap but would rarely think about the impacts of their purchase beyond their immediate selves. Sylvain Allard, a design professor at UQAM, highlights the difficulties with redesigning. Cartons, plastic bags, to jugs for milk, all have their pros and cons. The jugs may be more recyclable but can't block light which is harmful to milk. The plastic bags may be super cheap for milk but aren't very recyclable or strong. Finally cartons may be the best protectors but with that wax coating makes them impossible to recycle. Thus as far as the current modern society has progressed, EEQ proposes a shift from a consumer based culture to a waste mitigation or sustainable culture. This design trend has been very common in the 21st century as more and more companies not only use it to better the planet, but drive their stakeholders to get an additional boost of happiness or empathy when they see themselves as doing their part for the planet. So not only is it good for the earth but taps into a new market and strategy.

Part of the issue with packaging proposed by this documentary is a balance between overpackaging and under packaging. The goal is to protect the package but some designers take it to an extreme, overpackaging their products and thus creating unneeded substantial waste. This ties into the previous problem and thus the overarching problem of mitigation/balance. The goal will be a balance of strength, sustainability, recyclability, the product's decomposable nature, and it's image in the eye of the stakeholders.[5]

Key points for a good Product-Market Fit [1]:

- Production of product
- Preventing breakage during transport
- Slowing down the aging of products to prevent waste
- Protect user
- “Easy and agreeable method of handling”
- Useful and simple
- Ergonomic
- For shipping, it must be able to regroup objects of different elements or repeated units to reduce waste from overpackaging. (Ex: shipping a single spoon in each box versus 50 in one)

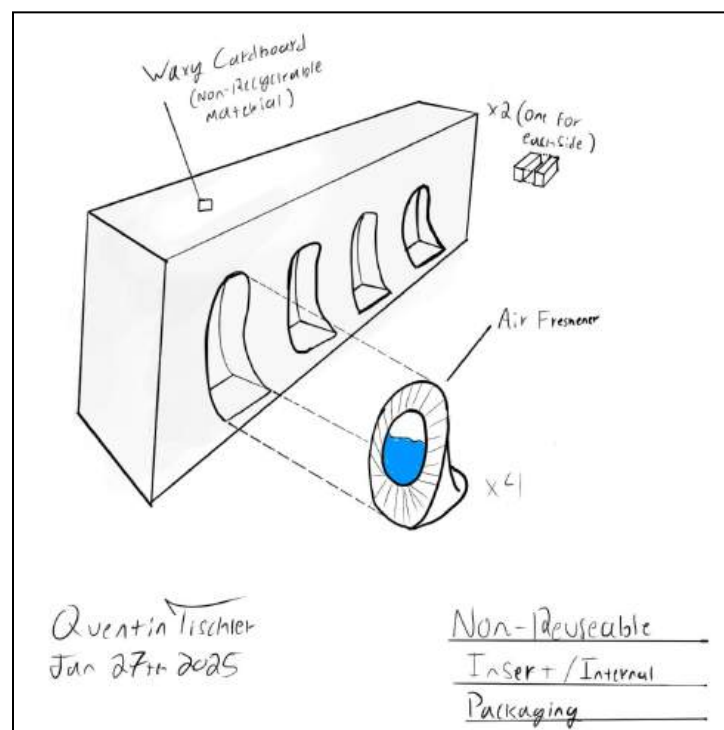


Figure 3

Figure 3 shows a common fitting design for the interior packaging of a product needing to be shipped. This one is a 2 part clamshell design but without the folding functionality.

Manufacturers encase the product, in this case an air freshener, in a cardboard form fit “sarcophagus”. The issue with this design is that it uses a wax coating over the cardboard, making it unable to be recycled with other standard cardboards. This shows a clear, simple gap in the market of sustainable design, which could be improved upon as well as the problem itself.

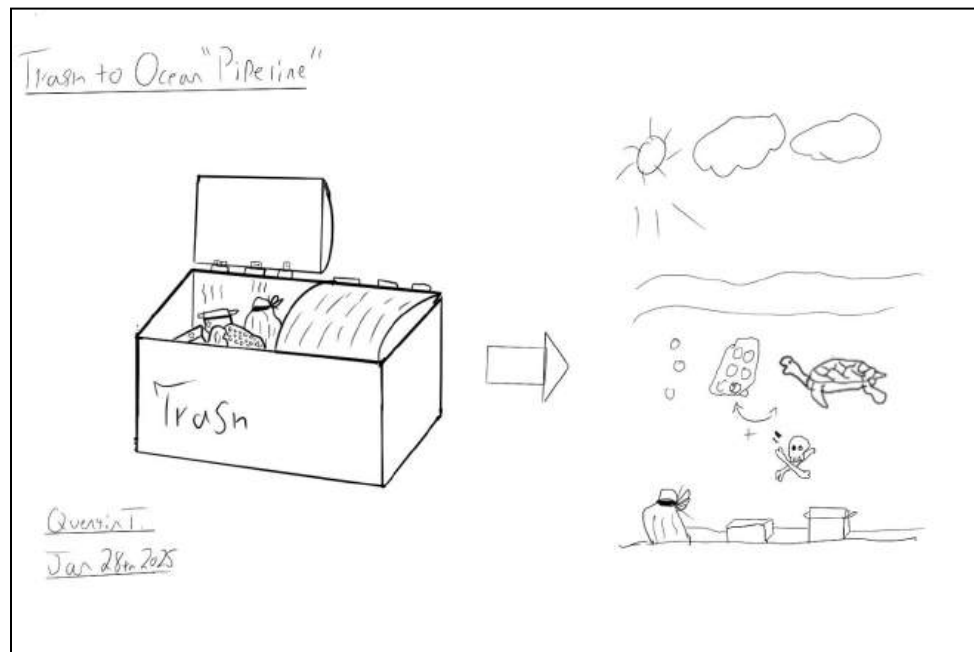


Figure 4

Figure 4 shows an additional part of the problem. A trash to ocean “pipeline” millions of pounds of trash entering the ocean each year it’s been seen that it not only affects the sea life such as turtles consuming harmful plastics, but also in humans with microplastics such as PFAS and other chemicals which health experts are apprehensive about entering drinking systems and other vital parts of human infrastructure due to their harmful properties

Interview Summary With Peter Frauenheim:

Peter Frauenheim is a 1st year college student who receives about a package a week of various items from LEGO to dry goods, wet goods, and more. When interviewing Peter, some important details came to light. First of all would be areas of waste in the packages he regularly and irregularly receives. One such item is Gatorade water. When bought in bulk, they come in a nonrecyclable thermoset plastic. This plastic, since it is “thermoset,” or unable to be remelted and shaped, while it is good for keeping the individual bottles together, is a massive waste of material and is nonrecyclable. Not only that, but it is similar in size and shape to other plastic matters that sea life are known to mistake for food and feed off of. Companies such as Thrive Market, which he receives goods from, ship their products in needlessly large packaging, causing waste. Had these been fragile items, he could understand the ends for overabundant packaging. But since they were mainly dry soups and goods that were already individually well wrapped, he thought it needlessly excessive.

An interesting point of reference he gave is that as a consumer, if it is a box that is made of cardboard, he will immediately put it into the recycle. Not looking at any numbers or any other indicators of whether that packaging is recyclable or not. It could be covered in a wax film or

any number of other things; it does not matter to him. This could be a result of a culture of recycling where, at a very young age, individuals are told to recycle, recycle, recycle, recycle, and so they recycle the materials they have been told about such as plastic and cardboard, never mind whether they actually can be reused or not. Recycling companies have been known to throw out entire batches of recycling if even one piece of recycling is not “up to code”.

When asking Peter what can keep him and other people he knows away from sustainable products, he referenced a few key factors. One being strength. If a box, paper straw, or any other sustainable item can't do its task as well or at all, then it will be forgotten. This may seem obvious at first, but too often, designers forgo effective products for merely sustainable ones. This may simply be a pro and con give-and-take, but it is important for designers to reasonably come as close to the original product as possible. Another key piece of failure he thought of was water resistance. Humidity and water are probably the most destructive factors when it comes to packaging and items or goods spoiling. Thus, whatever packaging item that is developed must be water resistant at the minimum. Last but not least, he thought of ease of use for the user. If he doesn't know which parts of the packaging are recyclable and which aren't, then he's probably going to either throw it all away or put it all in the recycle. With that in mind, he also said that another part of ease of use is how easy it is to open. Is it an intuitive folding design that has no tape, or a super hard box that you have to cut through to get into and may accidentally cut your product? These are many problems and concerns that he had with sustainable packaging and are useful considerations for this project.

Problem Area On-site Analysis:

Moving from room to room in Spruce Hall analyzing packages, their contents, their filling or interior packaging, and more yielded a good amount of quality information broken down into parts below.



Figures (from left to right): 5, 6, 7, 8, 9, 10

When it comes to interior packaging of a box or more form factor parts of packaging, there are a number of points that can be innovated from. The key problems here as viewed above are the plastic bags, the thermoset plastic wrap for the body armor(Figure 9), and the wax coated cardboard for the black air fresheners (Figure 8). These all are unrecyclable materials that have a long life in the environment from the point they are thrown out. The cologne sampler's tray(Figure10) is recyclable and thus shows what can be done with recyclable materials and is a good point of inspiration. The perforated tissue paper is a good substitute to plastic packing peanuts that was found during the investigation but is very thin and is not protective for the box it is inasmuch as it is just to fill the box. Moving away from just this recycled "filler", to a more sturdy recyclable, or made from recycled components filler, could be a great point of innovation for the team.



Figures (from left to right): 11, 12, 13, 14

Core boxes for packaging are a big part of recycling. As can be seen in the photo with the recycling bin (Figure 13) people throw all sorts of boxes and cartons in the trash, whether it's food that has not been washed out, or wax carton's which are not recyclable. This limits what can be used to recycle. Recycling companies tend to not accept food contaminated pieces or non recyclable wax cartons. So it is important that whatever packaging method is developed, the team needs to clearly define for the user what is and what isn't recyclable.

The three boxes shown here show a key part of waste mitigation for packaging. The balance of over packaging and under packaging. Here none suffer from underpacking but some do from over. The largest box (Figure 12), from Thrive Market, contained a variety of different dried good foods with the previously shown wrapping paper and plastic bags. While this is good that all the parts appear based on every indication to be recyclable, it is an abundant waste of resources to package such an item that's already well protected as dry goods are especially when they arrived with so much wasted space. The soap (Figure 11) and "snack filled" boxes (Figure 14) show an ideal minimum use of resources while maximum utilization of space and protection. By taking the wall in on the soap box and specifically form factoring it to the size, it allowed the designers to double the wall thickness while still staying under the size of say a regular amazon box. It is important to note that while it's very easy to be critical these boxes do have different jobs. The dry goods box might be for bulk purchases for consumers who purchase even more than this dorm individual did. Or the soap box since it is a specialty company can invest in custom sized/made boxes. But the point still stands that all this offers research towards producing a better less wasteful packaging method. Currently it's simply a battle of pros and cons.

Stakeholder Research:

Plastic is a major contributor to environmental trash/debris and subsequent issues. Of the 9.2 billion tons of plastic that has been reduced, at most 9% of it has been properly recycled [2]. Plastic isn't biodegradable like some forms of cardboard and other more sustainable options known today. This poses a large problem because plastic won't just go away in the environment. It will sit in the ground and degrade over time into smaller and smaller particles before eventually moving on to wreak havoc with another system such as potable drinking water. Single use plastics for packaging are the greatest offender of all. Their low user life span means they spend little residency with the user and predominantly spend their residency in a landfill which is counterproductive to the entire idea of sustainability. This means that a simple reduction in plastic quantities for packaging can't be the sole solution for designers. Either a complete material shift, or a solution in the form of a new recycling method which makes single use plastics no longer "single use", is the only viable way to tackle the problem of plastic packaging. The issue is recycling is expensive along with materials research. Disposal and labor cost for recycling are the key influencers here for why recycling is more difficult than the general public thinks[6]. In theory it is possible to use or recycle most materials in one way or another. The problem is the costs of time and money associated with them and the very little bit of business venture profit which is associated with them. Thus it becomes more and more difficult to recycle. Tying into that idea it is also generally tied together that the worse something is such as PFAS, the harder it is to get rid of or recycle. Creating a double negative scenario which doesn't cancel out, but compounds issues upon issues.

Even with non plastic based packaging such as cardboard, companies seem to use an over abundance of material leading to an increase in fossil fuels for more material production, including an overall increase in waste, and overconsumption of resources. Packsize, a company that prides itself on reducing packaging waste, says that "The average shipping box is 40-60% empty space." [3] That's 40-60% more wasted material than is necessary. Not only is this incentive bad for the environment, it's also bad for the industry. That's roughly 40-60% more cost additionally. This is clearly a problem that needs fixing. Tying into that idea of wasted space they have to fill it somehow so they either A over package the item so it can safely bounce around in the larger box, or B use more material and thus more waste to fill the empty space. Referencing back to the soap box above in Figure 11. This is a good and proper reduction of material waste which sustainable companies such as packsize are able to introduce to markets.

However not all companies have such regular recurring shapes or items. Plus the more different box sizes you have the greater the cost in the industry, if amazon for example made a box for every type of product they sold to perfectly fit the cost would be enormous. This is part of the issue of sustainable design. Deniz Turkcu and Nina Tura, two individuals at the Department of Industrial Engineering and Management at LUT University, identify numerous problems with sustainable design that have not regularly been acknowledged or known by the general public. These issues include but are not limited to "high investment and operating costs, lacking standards and guidelines, varying regulations among countries and regions, changing regulations, lack of knowledge of policy makers, consumers and end of life management companies are found to be causing most of the tensions." [4] These issues range from the product itself to the industries marketing, creating, and shipping these sustainable ideas. It's important for designers to understand these facets of sustainability in order to fully comprehend the net change their

product brings about. A comparable idea would be, what's the point in an electric vehicle if burning fossil fuels to create electricity for the car creates more CO₂ than just running a gas car. These problems can be mitigated in various ways but are still part of the core process.

1.7.2.2 - Remaining Unknowns

The key lapse in information from the research this section is focused on is materials research. This is something excluded from this research portion because it is more tightly linked with the specific chosen redesign that the team will ultimately decide on. A good example would be if the research above focused on compostable papers for filler, but instead, the team focused on food packaging. Compostable paper would not be a good choice. Thus, the remaining unknown is narrowed down to specific materials. If it is needed to expand further into unknowns, one could look at each recycling plant and see which materials are commonly recycled and which aren't, but once again, that is more tightly linked with Criterion B or the planning phase of a project the overall background. It does no good just to list out everything that can and can't be recycled.

1.7.2.3 - Summary

Packaging is not only a huge pollutant factor to the ecosystems but also a waste of money and resources, which are either non-renewable or take long enough to renew that it isn't feasible to consume them at the current rate without an alternative. Coupling that with their long lifespans and the fact plastic will never decompose means that a solution to waste management pre and post-product use needs to be taken into account. It must be an acceptable enough substitute that the population and manufacturing industries or a more specific target audience is willing to invest in the switch. Sustainable packaging isn't just a net gain, there are losses such as capital, new pollutants from manufacturing, lack of standardization, and more. There are many different places for improvement such as food packaging, filler mitigation, an alternative to filler, and more that can be addressed. What it comes down to is narrowing the problem and developing a solution for the team.

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1.7.3 - Jack Morris

1.7.3.1 - Research Summary

As a part of everyday life, packaging contributes a significant portion of waste as it is used in almost every product someone buys. With the rise of E-commerce and other similar fields, the need for single-use packaging has increased dramatically, and the waste that it produces is a problem that only keeps on growing. One example of this problem is, according to the article “Product Packaging by E-commerce Platforms,” India, which does not have enough regulations on waste, creates about 10% of all global waste, which massively contributes to the microplastics that make their way into all ecosystems across the world. Not only that, but the rotting packaging releases CO₂ into the air, which contributes to global warming and also destroys ecosystems [1]. Creating a more sustainable form of packaging that can be used on a day-to-day basis will be crucial to help reduce these environmental problems.

After viewing “Unwrapping - the documentary” by Éco Entreprises Québec, waste due to packaging starts from the design of the package and is less about how one uses it. As the world has turned into a more consumer-heavy market, the design of current packaging makes everything feel like disposing of all packaging is the norm. Transitioning into a world where the norm of packaging is the three R’s- reduce, reuse, and recycle- is crucial to turning the world into a greener place. Most of the recyclable materials are not properly recycled due to natural habits of throwing away a temporary item. For example, if someone is given a compostable coffee cup at a cafe, the cup is likely to end up in a landfill instead of a composting facility since the person assumed the cup was non-compostable. If any packaging is meant to be recycled, more design elements should be added so that the user knows how to properly dispose of the item. Other issues that lead to large amounts of waste is overpackaging, as some products might have too much packaging for what they're worth. On top of that, many products are a part of secondary and tertiary packaging to ship the product from the factory to the store. For instance, a bottle of sunscreen might be packaged in groups held together by plastic, then those groups are placed in boxes and put on a pallet, and finally, the pallet is wrapped in another layer of plastic as seen in Figure 5. Adding all of these extra layers created even more waste on the back end that most customers do not see [2].

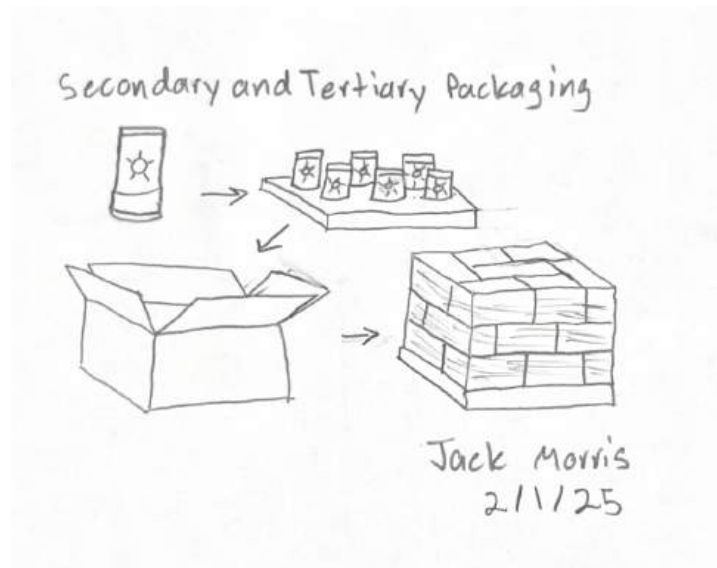


Figure 5: Demonstrates how certain products use more packaging when shipped in large quantities.

One big problem that arises with sustainable packaging is choosing the right material to use to both protect the product inside the package and minimize waste. According to the article “Need for Sustainable Packaging: An Overview” by I. D. Ibrahim, “plastic-based materials are the most used packaging materials” as it is cheap to mass-produce. Plastic will create many problems, including people not properly recycling it and having plastic accumulate in landfills. Finding ways to recycle plastic is incredibly difficult as, right now, many plastic products are “too complex or too small to recycle.” [3]. Another primary material used is cardboard or paper, as they are both easy to handle and cheap to produce, but they do not handle water and harmful chemicals very well. Ultimately, all of these different products need to have one common goal: the ability for people to easily recycle or reuse these materials [3].

After interviewing Gregory A. Morris, CEO of Nexcopy Inc., he brought up many issues that occurred in modern day packaging such as expense and use of non-recyclables. One of these problems included how he needed to ship out one of his main products, known as the Duplicator, using a custom-made foam material to fit the dimensions of the product, as seen in Figure 6. Multiple problems arise with this foam material, namely the cost, as it costs him \$16 per Duplicator to have the custom foam produced and shipped to his warehouse. Another problem with this is that the foam is nonrecyclable, and since it is custom-made, it will not be reused when it reaches the end user. Shipping out many of these products a day, this style of packaging gets both expensive and produces a large amount of non-reusable waste. Some other widely used packaging in this workspace include bubble-lined manila envelopes, used for shipping flash drives, and plastic bagging to wrap other products. Talk about potential solutions also arose during the interview. In the past, Morris has tried using cardboard paper to protect the product, but it needs too much paper, and too much time goes into wrapping the product, so just using the foam packaging was the better solution. One example of a solution that he thought about but had not tried is to use a 3D printed mold and some spray-on foam to produce the correct shape to properly package the product. The only problem with this is that he had not found a sturdy enough foam to both protect the Duplicator and not have it move around inside of the box. This

information will help the team understand what smaller businesses would look for when attempting to make a more sustainable form of packaging [4].

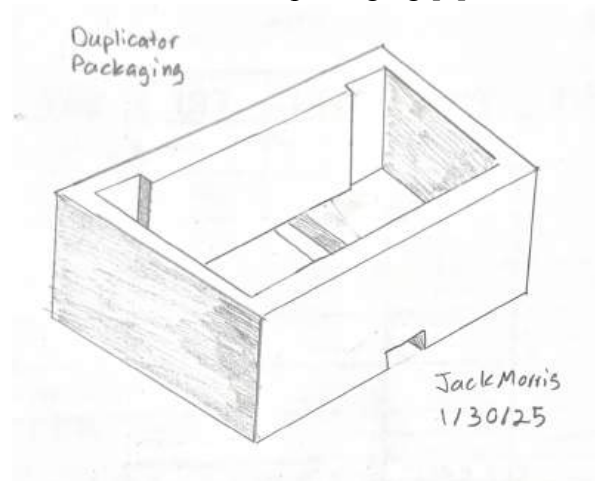


Figure 6: Shows one of the two pieces of packing foam needed to insulate and protect the flash drive Duplicator.

1.7.3.2 - Remaining Unknowns

After conducting interviews and researching topics related to packaging, there are still some unanswered questions that need to be addressed. One of these questions is whether it's the box that is producing the most harmful waste or if it is the insulating and protective material inside the box that causes more waste. Another item for the team to ask itself is, out of all the categories of packaging, which ones should be focused on? For example, the primary goal is to reduce overall waste, but should reusability be the secondary focus, or should recyclability be the secondary focus? Lastly, the team would have to consider the cost of manufacturing the packaging because cost is one of the primary concerns when it comes to packaging. Since packaging is needed in almost every market, if the cost of making the packaging is slightly more expensive, that additional cost will be multiplied by every single package that was used in that market. However, balancing all of these different issues will be difficult as the team will have to focus on some issues and not focus so hard on other issues.

1.7.3.3 - Summary

For an essential part of everyday life, the need for packaging is ever growing, and with the motivation for a greener world, sustainable packaging is at an all-time high in demand. With many different types of material in mind that can be recycled and reused, making the right choice of material will be vital for this team's success in making a more sustainable product. Adding on, designing the packaging in such a way that it is comparable in price to the other cheaper, non-recyclable counterparts will be difficult as sustainable products tend to be more expensive. Lastly, there are a multitude of categories that packaging has to take into consideration, such as usability, ergonomics, cost, aesthetics, regulations, and many more. With so many categories that packaging has to fit under, there can only be so many topics that the team can cover. Putting all of this information in mind will be vital for designing a product that can reduce waste and benefit

in other ways compared to the widespread use of non-recyclable materials that most companies use today.

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1.7.4 - Strydr Silverberg

1.7.4.1 - Stakeholder Research Summary

Packaging solutions are arguably more important than ever, with over 7.7 billion shipments being sent out by Amazon in 2021 alone[3]. The need for packaging products is only growing, and a key problem faced in today's society is what to do with the packaging once the user is finished. I spoke with Katzke Packaging Company's President and Owner Glenn Suta, who began his career with Katzke in 1984 as a sales representative[2].

Katzke Packaging specializes in packaging products and materials, with corrugated boxes and protective packaging being the most popular. According to Suta, "Boxes are interesting because each one has a specific use case. One of the most popular products is a single-walled corrugated box because of the low cost and ability to be used for a wide range of products. Triple walled corrugated, though, is only purchased by companies selling very large consumer goods like refrigerators or washing machines." While packaging solutions like triple-walled corrugated boxes aren't exactly common, they represent a large portion of the unrecycled cardboard.

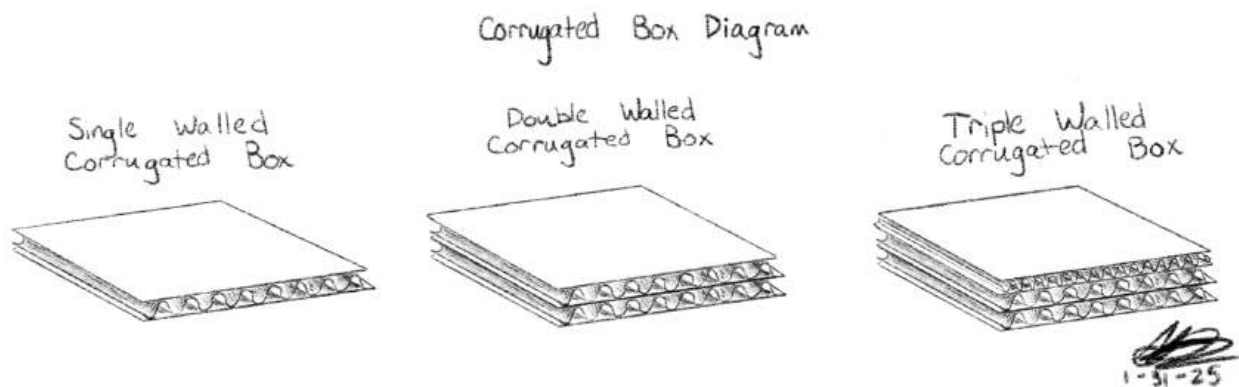


Figure 7: Shows different types of corrugated cardboard boxes.

One problem laid out by Suta was that cardboard isn't being recycled enough. According to the Environmental Protection Agency (EPA), approximately 80% of cardboard packaging is recycled [1]. While this number sounds high in theory, cardboard recycling is different because of the materials used to construct cardboard. Each new piece of cardboard can be recycled approximately 8 times before the fibers become too weak to reuse. This makes cardboard recycling especially important because it is such a well-defined process that it can be done economically, whereas many forms of recycling simply aren't viable due to cost.

One question that came from this conversation was if Suta thought that recycling of corrugated cardboard could be further optimized or improved upon. His response was, "I think that cardboard could benefit a lot from a look at the recycling process. We work with several corrugated recycling companies, and some major inefficiencies are clear to me. We sometimes struggle with communicating proper requirements for recycled corrugated materials that we are receiving, and this has caused us some quality issues in the past."

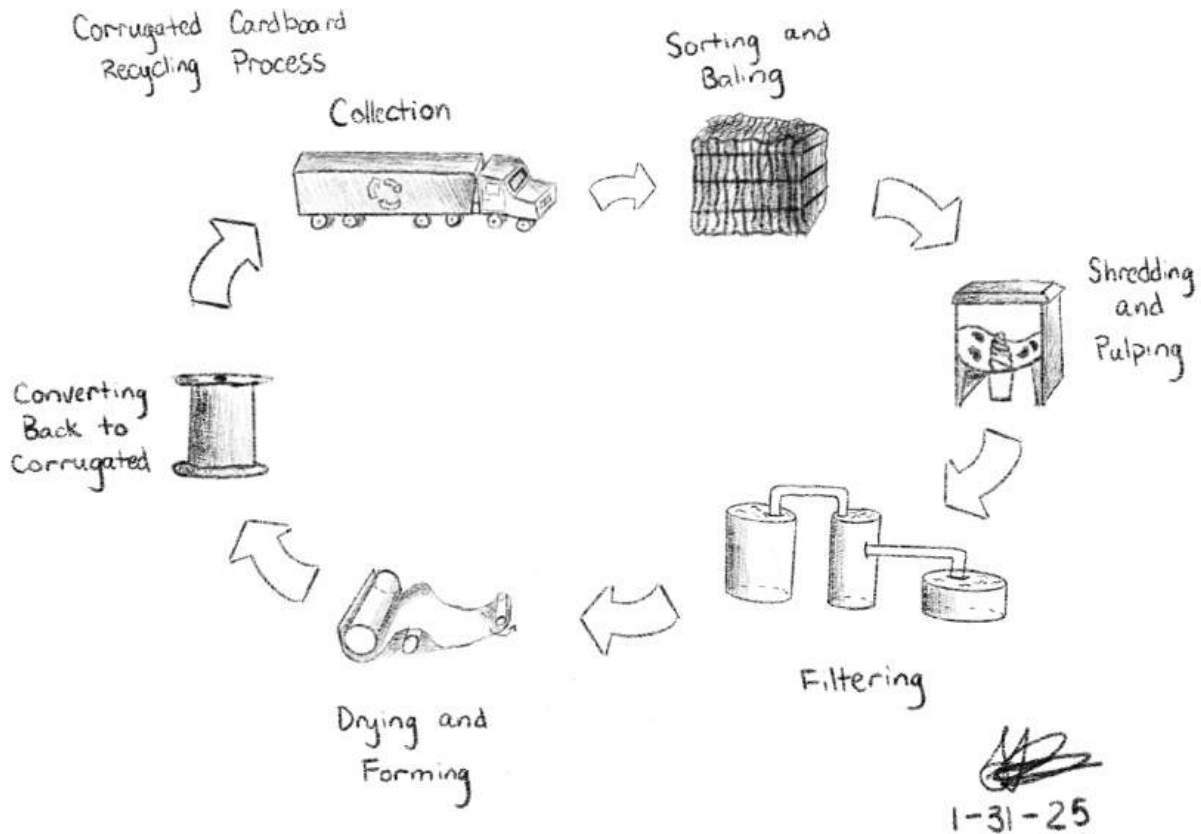


Figure 8: The corrugated cardboard recycling process.

Furthermore, Suta talked about potential improvements to existing packaging solutions. Suta said, "One suggestion could be to try and figure out a way to implement existing 'space fillers' with more sustainable products like cardboard." This idea is exciting because the corrugated recycling industry is one of the strongest and most effective, meaning that making a shift from something like plastic or styrofoam to something like corrugated cardboard would be impactful. According to the EPA, only approximately 8-9% of plastic packaging is recycled, compared to over 80% of corrugated cardboard [1]. This change could potentially be very productive if it is adopted by major packaging manufacturers and distributors.

1.7.4.2 - Remaining Unknowns

Despite meeting with an industry expert and getting some excellent starting points, some unknowns still exist relating to the topic of reducing packaging waste. The problem space is incredibly wide right now, and while that's a good thing, the field of packaging is vast, and the specific problem that needs to be analyzed has to be chosen. It would be useful to analyze some data relating to a few different problems in space in an attempt to analyze what problem stands to have the most reward from being solved. As mentioned in the Stakeholder Research Summary, attempting to solve supply chain issues related to packaging and its recycling could also be a potential path forward.

1.7.4.3 - Summary

The field of packaging, especially when related to corrugated cardboard, is vast and affects all lives every day. The choices that are made regarding packaging today could have drastic consequences on the future, making it an increasingly important question to try and answer. Packaging companies are becoming more and more interested in developing new and innovative solutions to the problems posed by modern packaging, retail, and e-commerce. Some potential options for this project include attempting to improve the way packaging and recycling companies interoperate and improving so-called ‘space fillers’.

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1.7.5 - John Davis

1.7.5.1 - Research Summary

Documentary Analysis for “Frontline: Plastic Wars” by PBS:

Plastic within society is one of the biggest problems that civilization faces today. However, the video “Frontline: Plastic Wars” sheds light on what the biggest issue within the plastic industry is. Out of all of the plastics in the world, only 10 % is recycled the way that consumers in America picture it. As it turns out, the amount of recycling done in the world is not driven by how much the consumers put into specially marked bins or containers but by the profitability of the plastics themselves. Plastic is a broad term that encompasses a wide variety of individual plastics with different chemical compounds. Certain products, such as milk jugs, can easily be melted down and made into little balls that are used to create new products. The low amount of resources needed to modify these specific plastics makes recycling a profitable industry for those that take in these specific types. However, most of the plastics that circle within the world take a lot more effort and resources to recycle, making them economically detrimental and unable to be sustained based on profits alone.

This massive inequality and lack of sustainability were first noticed in the 1970s, which led to massive riots and calls to change within the producers of these plastics. However, instead of these major companies changing their ways, which would take away a large portion of their profits, they decided to focus on advertising the benefits of plastics. This raised the public opinion of plastics and made the public forget about the vast majority of plastics that were harming the environment. However, this specific type of advertising only worked for so long, and public unrest began to rise in the 1980s. Instead of getting caught up in a series of riots and boycotts, plastic producers decided to implement a new system: the recycling symbol with a specific number or letters on their products. This system required all products that could be recycled to have this symbol, along with a number or series of letters to signify which type of recycling it qualified for, as seen in Figure 9.

						
PETE Polyethylene Terephthalate	HDPE High-Density Polyethylene	PVC Polyvinyl Chloride	LDPE Low-Density Polyethylene	PP Polypropylene	PS Polystyrene (Styrofoam)	OTHER Miscellaneous Plastics
HOW TO RECYCLE						
						
Curbside Recycling	Curbside Recycling	Difficult To Recycle	Outside Recycling Facilities	Curbside Recycling	Difficult To Recycle	Difficult To Recycle
PRODUCTS MADE FROM MATERIAL						
						
soda bottles, water bottles, salad dressing containers, peanut butter containers	milk jugs, household cleaner containers, cereal box liners, yogurt containers	food wrap, plumbing pipes, window cleaner bottles, detergent bottles	squeeze bottles, plastic shopping bags, carpets, food wraps	straws, ketchup bottles, syrup bottles, medicine bottles	compact disk cases, egg cartons, meat trays, disposable dinnerware	sunglasses, computer cases, baby bottles, bullet-proof materials

Figure 9: [1]

However, this system was widely misunderstood by their consumers because consumers began to believe that anything with the triangle symbol meant that it would be recycled completely. What the producers failed to mention was that while their products technically could be recycled, the numbers signified to the recycling companies which products would allow a profit and which would only lead to a loss in profits. Then, the producing plastic companies released a new series of ads that placed the consumer at fault, stating that it was the consumers duty to recycle anything that had this triangle symbol, even though most of the products weren't going to be recycled by these recycling companies based on the little number in the center of the triangle. The idea that all the plastics in America were being recycled completely in the 1990s was only reassured once China began to buy America's plastics to recycle them and turn a profit in the 2000s.

In reality, China would buy large bundles of American plastics, filter through to find the easily recyclable plastics, such as milk jugs, and then leave the rest of the harder plastics to be melted, releasing harmful chemicals into the air. Eventually, even China couldn't keep up with this system and stopped buying American plastics in 2018, leading the American public to find out what had been happening to their plastics in China and causing yet another uproar calling for more plastic regulations. Instead of changing their ways, plastic producers found another country to buy America's plastics, Indonesia, and began another ad campaign like the ones in the 1980s that placed the consumer at fault and claimed that the consumers just needed to keep doing what they have always been doing.

However, unlike in the 1980s, the American public is a lot more aware of the dangers of plastics, and a few of these major companies have begun to research new ways to recycle plastics, leading to new inventions being created. Despite these seemingly new advancements, it was discovered that some of these inventions were actually created and had already failed in the markets in the late 1990s, leading to yet another outrage at just how truthful the plastics industry is being with their consumers [2].

On-site Interaction with Consumers Completed Through Arkins Park Stone:

To find a more personal aspect on how packaging is wastefully used within the workforces, a special hands-on investigation was carried out at one of the most unlikely places possible, a rock quarry and rock refinement business. This business, Arkins Park Stone, drills the rocks used directly from the mountain their property is on, cuts them, and turns them into modern-day countertops, pedestals for statues, aesthetically pleasing guards for the sides of major roadways, and a variety of other rock-based needs. While this may give the impression that this company only uses rocks and saws, with maybe a chisel or two in the meantime, they use lots of plastics and other man-made materials in their craft. At the beginning of the process, usually dynamite, drills, and massive forklifts are used to gather the rocks needed and place them where they need to go. Then, saws are used to cut the rocks into their needed shapes and sizes. Then, the rocks are sanded, packaged, and shipped out to wherever they need to go.

This investigation took place in only part of this process, which consisted of the cutting, sanding, and packaging of these rocks. The cutting of this process is carried out by replaceable saw blade tips, which are manually welded and come in foam packaging, as do the large saws themselves. These saw blades and tips don't need to be replaced often, though. Only the tips need to be replaced every couple of months to ensure clean cuts. Once the rocks are cut, they are sanded to the customers' specifications using disposable pads. After sanding, the rocks are packaged using a variety of cushioning elements. Workers place a foam corner on the rock's top corners to avoid chipping during transport, and then the rocks are wrapped with heavy-duty saran wrap, as seen in figure 10.

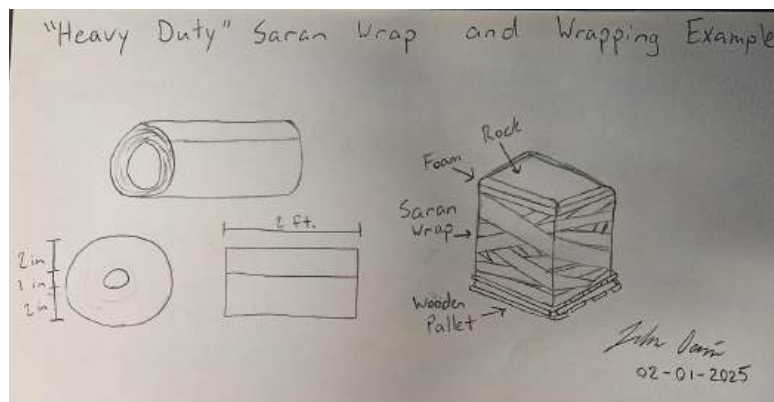


Figure 10

Despite this saran wrap being supposedly thicker than the kind commonly used in kitchens, it can very easily be ripped through by the rocks' uncovered edges, leading to many wrappings being needed. This specific process uses lots of plastic and foam to ensure the rocks' safe delivery but includes lots of waste once the rock is delivered because the wrap cannot be reused. The wooden pallets that the rocks sit on are large in number and could potentially be a problem in their packaging, but a sustainable system of pallet reuse and money back for returning them after delivery has successfully combatted this risk [3].

Mock Consumer Standpoint:

Following this, a consumer's interaction with packaging was created to understand how much excess packaging is used. For this interaction, a speaker was delivered by Amazon shipping. Despite only ordering a single speaker, two boxes, wrapping paper, masking tape, plastic

placeholders for the product, and a coated instruction manual were delivered, as seen in Figure 11.

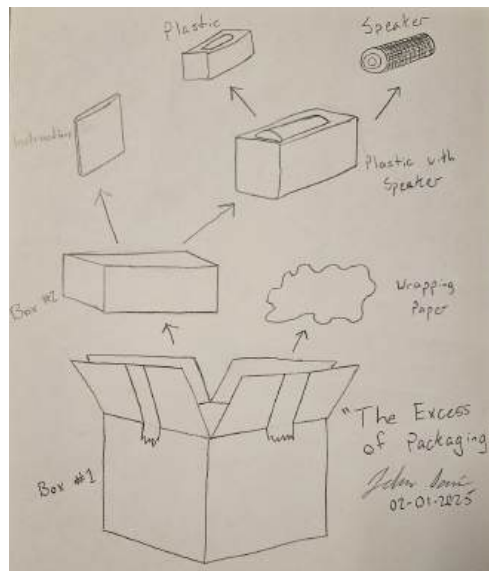


Figure 11

Based upon this, it would be probable to assume that most packages delivered by Amazon, one of the largest shipping companies and online stores in the world, also contain the same amount of excess packaging as seen in this delivery. While the importance of securing a package and making sure its contents are safely delivered are a crucial matter in the world of online shopping, it seems that there is a far greater need for increased materials than before the invention of online shopping, creating an even greater need for an improved recycling system [4].

1.7.5.2 - Remaining Unknowns

While extensive overall research has been done, certain questions remain unanswered. One such question is whether the box needs to be changed or if the insides themselves need to be changed. Another question is what drives the current state of packaging and its innovation or lack thereof. Another is what kind of sustainable materials can be used in place of plastics and other materials commonly used in packaging today.

1.7.5.3 - Summary

Plastic usage and excess packaging materials are both past and present problems. Plastic has been a problem for over 50 years and has been actively hidden by major companies through ads, misleading symbols, and shady business deals. Packaging materials are a part of everyday life in multiple businesses across the world and are often used excessively to protect their products. However, once received by the consumer, lots of this packaging is tossed out, leading to excessive trash and waste materials.

1.7.5.4 - Individual Bibliography

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[3] On-site participation with Arkins Park Stone carried out by John Davis on 02/02/2025

[4] Consumer standpoint carried out by John Davis on 01/26/2025

2.0 Existing Solutions, Alternatives Reviewed, and Option Selected

2.1.1 SUMMARY OF KEY STAKEHOLDERS

Nearly every person and every company needs to ship products to and from their respective locations. Either someone ordering a product online or the companies who need to ship products to their consumers. Some of the key stakeholders would include a very wide range of people and companies. One of the stakeholders is packaging manufacturers, as they would have to produce this product many times per day, so reducing the cost of production and making the production process more efficient would be imperative. Another stakeholder would be shipping and logistics companies and other companies that need to ship out products as they would have to handle these products day in and day out. Adding on, the consumers that receive these packages would have to know how to handle the box, such as how to open it properly and how to recycle it properly. Lastly, environmental regulatory bodies would be key stakeholders in making a more sustainable product that is as versatile as packaging. This product must be sustainable. To conclude, each of these stakeholders will help the team with feedback on what they need and want out of this product.

2.1.2 EXISTING ALTERNATIVES AND SOLUTIONS

Many current products exist that attempt to solve the problem of reducing product waste in packaging while ensuring product safety. Some of these include Wood Chips/Fine Excelsior, Compostable Bubble wrap, compartmentalized boxes, and more. These three are common examples and have their pros but also several cons. Starting with FE or Wood Chips, it is made from aspen and has the benefit of being dust-free and hypoallergenic. But it does not protect to the same degree bubble wrap or other fillers do. It additionally makes a mess when opened and is nonrecyclable. But it is compostable, which is a plus. Another would be compartmentalized boxes with reduced sizing. This is good for items such as wine bottles and other already sturdy glassware. It is also recyclable. But unfortunately, it is very unit-specific and relies on a great deal of the strength of the glassware itself, not necessarily protecting it. As a final plus it is cheap. Last but not least is compostable bubble wrap. Its main pro is being recycled out of materials and compostable. Unfortunately, it also requires a fair amount of material and is very expensive (\$40 compostable vs \$12 normal) relative to similar products.

Overall, each of these existing products has its pros and cons. Leaving room for improvement in multiple areas for re-design and/or innovation. Such as versatility for products, recyclability, cost, ease of use for users, product cleanness, and more.

2.2 REQUIREMENTS, CUSTOMER NEEDS, AND TECHNICAL SPECIFICATIONS

The redesigned packaging must balance longevity, reusability, cost, protection, ease of use, versatility, production feasibility, postal requirements, weight, and user perception. To ensure

longevity, material research and decomposition estimates will determine if the package lasts between two weeks and two years, with potential water tests for degradation analysis. Reusability will be assessed by evaluating whether components can be reused or recycled and how many times the packaging can be effectively repurposed. Cost estimation will rely on publicly available material pricing to calculate the production expense per unit. Protection testing will involve drop tests from various heights to ensure the package maintains its structure and safeguards its contents. User ease-of-use will be examined through hands-on testing, allowing individuals to interact with the package to determine if it can be intuitively opened without damaging the product. Similarly, the producer's ease of use will be evaluated by assessing the time and complexity of assembling and packing the product. Versatility testing will include trials with different product shapes and sizes to verify that the packaging meets the needs of multiple industries. The feasibility of production will be determined by ensuring that no custom tooling is required for manufacturing. Compliance with postal requirements will be confirmed through research, verifying that the package meets shipping standards. Weight must remain under two pounds, which will be confirmed through direct measurement. Lastly, user perception will be evaluated via surveys to gauge reactions, ensuring the design is visually and functionally appealing without causing confusion or dissatisfaction.

2.3 INDIVIDUAL LOOKS-LIKE PROTOTYPE AND CONCEPT

2.3.1 - Textured Double-Walled Box with Circular Insert

2.3.1.1 - Description

This is a double-walled corrugated box with a rectangular grid cut into the top layer of each wall of the box, adding texture, improving aesthetics, and improving user grip. It has been reduced to the minimum size to fit the product, ensuring a tight, secure fit. The circular insert is there not only to add additional protection to the inside of the glassware such as this box is modeled after (4-cup Pyrex) but also to appease consumer expectations of filler. The reduced size, along with benign recycled cardboard, allows for double walling with sustainability in mind to stay eco-friendly and increase the protection of glassware. Compostable tape is eco-friendly, and the box has intuitive markings for easy deconstruction and subsequent recycling.

2.3.1.2 - Field Sketch

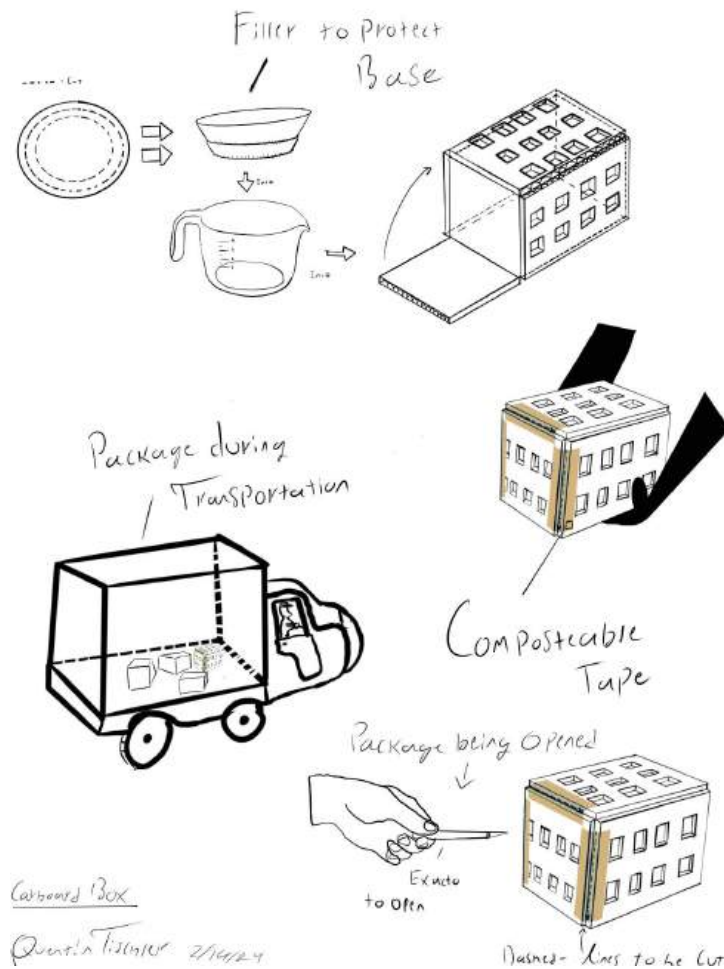


Figure 12: [Here, the product is seen as its real-world application. One part includes the box being transported in a truck along with other packaging of different items. A user holding and gripping the box can be seen along with another using an Exacto to open and subsequently deconstruct the box for recycling. At the top is a 4-cup Pyrex, its circular insert, and the box open, ready to be fitted with the glassware and then closed and taped before being transported to the truck and then to the user.]

2.3.1.3 - Looks-Like Prototype



Figures 13,14,15 (left to right in order):

[Figures 13, 14, and 15 show the core box, its double-walled features, and its rectangular grid. Additionally, the compostable tape is visible along with their marked lines, showing users where to cut to deconstruct the box for easy recycling.]



Figures 16,17,18 (left to right in order):

[Figures 16, 17, and 18 show the circular insert. The top is shown in Figure 16 with its 4 full-depth cuts. Figure 17 shows the bottom with its 2 main circular ½ depth cuts and 16 half-depth relief cuts for folding. Figure 18 shows the looks like ½ scale fitting perfectly halfway into a full-scale pyrex, showing it's been properly dimensioned.]

2.3.1.4 - Solution to Problem

This design offers several solutions to different current problems with the packaging of fragile items. One such issue is over-sizing of packaging and thus a need for an increased amount of filler to fill “dead space” in the packaging. By reducing the overall size of the box to just barely (1mm>) larger than the dimensions of the product (for this looks like it was a 4-cup pyrex), it dramatically cuts down on used storage. The use of recycled cardboard increases the box’s sustainability value. Since the overall size has been cut down, the material can be redistributed to the walls of the box, adding a second layer of corrugation to protect the fragile product inside

even more. Rectangular cuts in the walls, top, and bottom add additional points of grip for the user to utilize, which reduces the chance of falling or dropping in the first place. It also adds additional texturing and aesthetic appeal to the audience, which consumers expect when purchasing expensive or higher-end items such as glassware. The box is held together by 6 pieces of compostable tape, which have markings showing where the user can cut to deconstruct the box. Deconstructing the box limits space in the recycling bin, meaning consumers are more likely to recycle more or not have to take out the bin more often. A circular insert with 4 full-depth relief cuts, 2 main circular $\frac{1}{2}$ depth cuts, and 16 half-depth relief cuts allows the insert to be placed flush at the bottom of the pyrex. This not only provides additional protection for the product's insides but also appeases the common consumer expectation of some sort of filling.

2.3.2 - Inflatable Cushion Inserts

2.3.2.1 - Description

Instead of focusing on making the product more recyclable, this prototype focuses more on the reusability aspect of packaging. First, the shape of the box has only one lid that closes, so the user only has to tape one side of the box instead of four or more parts, as seen in Figure 19. Another feature of the box is the cushioning inserts to protect the item that can be inflated and deflated to multiple levels, depending on the size of the product. This means that the box can be used a multitude of times before it has to be thrown away, like many other forms of packaging that can only be used one time for one type of product.

2.3.2.2 - Field Sketch

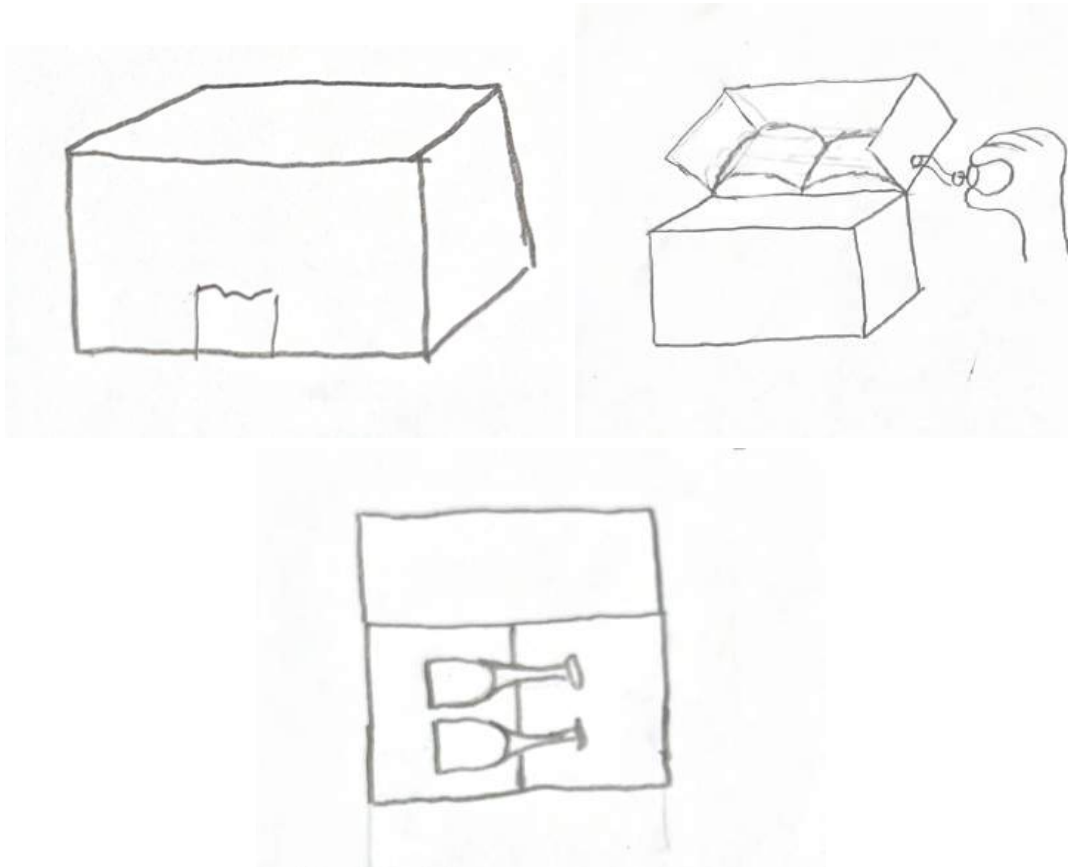
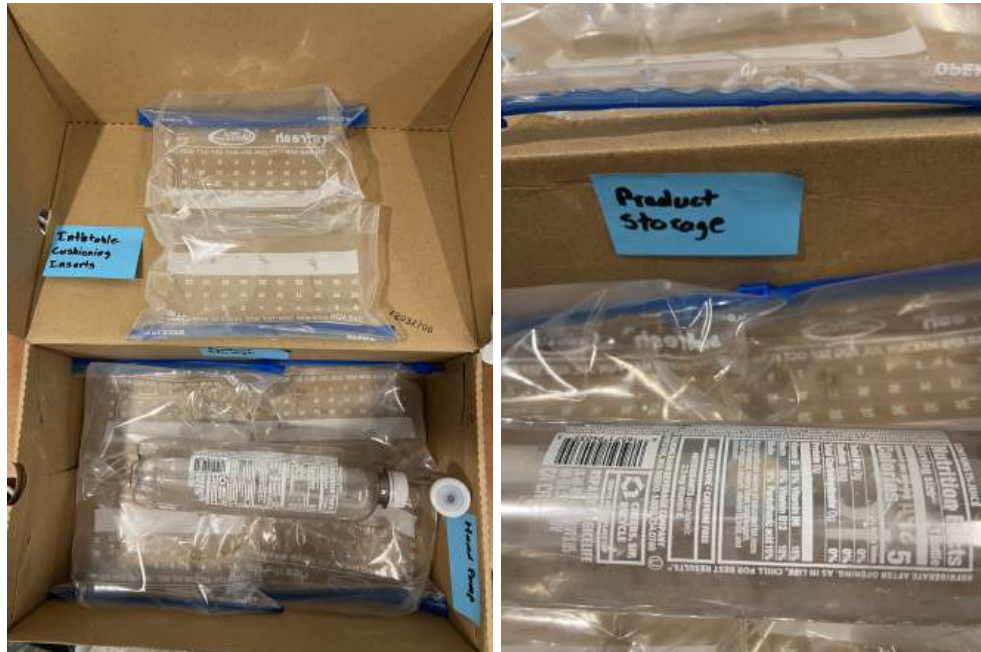


Figure 19, 20, 21: [These figures show the packaging in use of storing a fragile item. They do not include dimensions as these boxes can be scaled to any desirable size depending on what the user wants.]

2.3.2.3 - Looks-Like Prototype



Figures 22, 23: [These two figures show what the product might look like in the real world. whereas in Figure 22, the bottle cap represents the hand pump that will inflate the inserts.]

2.3.2.4 - Solution to Problem

The inflatable cushion inserts can protect the fragile products that are being shipped inside the packaging, as shown in Figure 23. These inserts can also be custom-inflated to the right amount to make sure that the packaging can fit any item that can fit in the box. Adding on, since the inserts can be deflated and inflated, they can be reused many times, reducing the overall amount of waste that comes with packaging, as seen in Figure 22. The reusability aspect can also reduce costs for the packaging as there will not as many boxes need to be produced as they can get passed around over and over again. All in all, this design dabbles in many different aspects, such as reducing waste and cost for the package.

2.3.3 - Dissolvable in Water Mesh

2.3.3.1 - Description

Rather than focusing on the cost of this issue, this prototype focuses on the sustainability and safety of fragile glassware packaging instead. This specific prototype is meant for glass bottles, such as wine bottles, that are both extremely fragile and contain a fluid that must stay safe during its travels. As shown in Figures 25-27, this specific design features a mesh on the bottom of the box that acts as the primary cushion for the bottle and is made up of dissolvable string, such as that found in dissolvable stitches used after oral surgery. This material would allow not only the safety of your bottle through its proven strength and elasticity but would also degrade over time,

preventing it from having a lasting impact on the environment. To keep the bottle in place and firmly against the mesh, the design also features a plastic clasp for the quark to rest in and a cardboard cutout for the base of the bottle, as seen in figure 24. To ensure the stability of the box and mesh connections, cardboard inserts are placed over the sides of the mesh, providing extra protection. Currently on the market, protection for glass bottles is not only extremely inefficient but also damaging to the environment. Many use copious amounts of styrofoam, plastic, or shavings to keep their product safe, which can cause lasting impacts on the land. In contrast, this design minimizes environmental impact and materials used without compromising product safety.

2.3.3.2 - Field Sketch

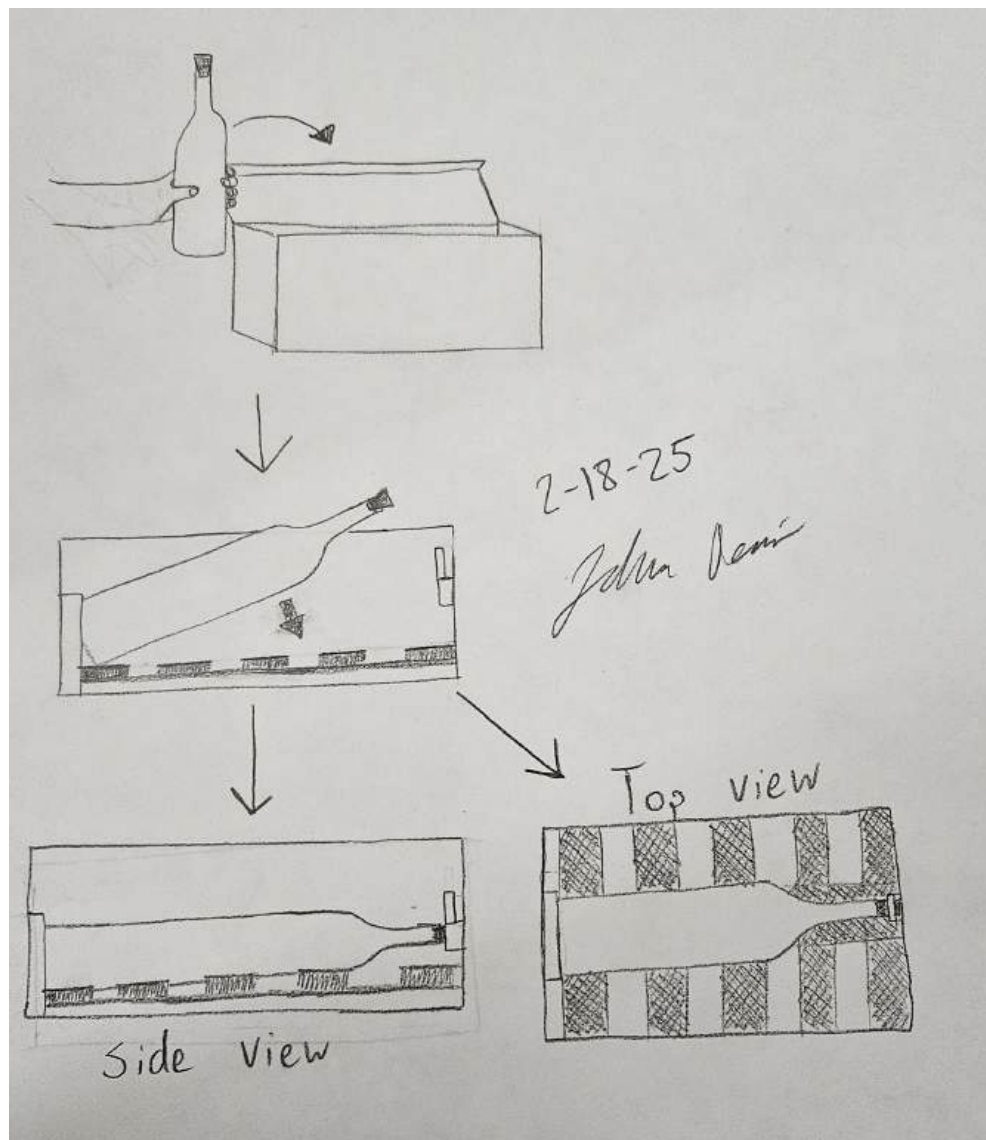
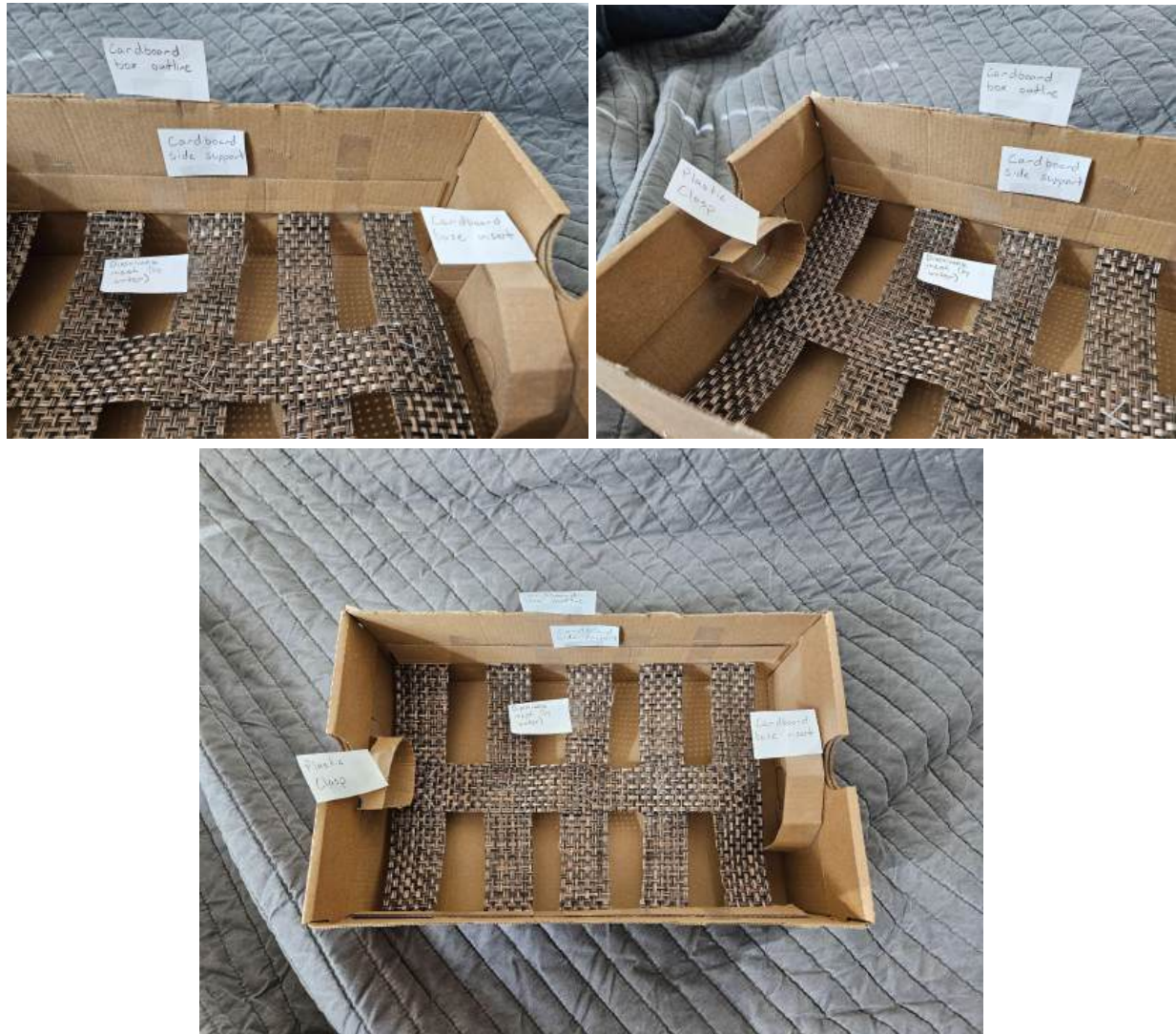


Figure 24: Visual Representation of How the Design Would Be Used

2.3.3.3 - Looks-Like Prototype



Figures 25, 26, and 27 (left, right, and bottom): Pictures of the Prototype from Various Angles
2.3.3.4 - Solution to Problem

This specific design solves not only the environmental challenges of past designs but also the potential product safety of the bottles. By creating a pressure absorbing mesh and locking the bottle in place with the inserts, the bottle is protected from most impacts from any direction. By making the mesh dissolvable, the box becomes much more eco-friendly and supportive of the environment by creating a simple way to safely dispose of the packaging waste after use. The simplicity of this design also helps support the packagers and manufacturers because it doesn't require a lot of specialized tools aside from making the mesh itself. This makes this product's creation, assembly, and usage both straightforward and effective. Overall, this design is not only better for the packagers and manufacturers, but also the customers and environment as a whole.

2.3.4 - Cardboard Mesh Filler

2.3.4.1 - Description

Cardboard mesh filler is a flexible cardboard material with cuts across the entirety of it to allow it to flex to fit a variety of products. My product was inspired by existing solutions like crinkle paper filler and other cardboard inserts that aim to protect the product. My product is an improvement on these existing solutions because it is producible with existing manufacturing techniques and machines, as well as provides adequate protection during shipping for moderately fragile products.

2.3.4.2 - Field Sketch

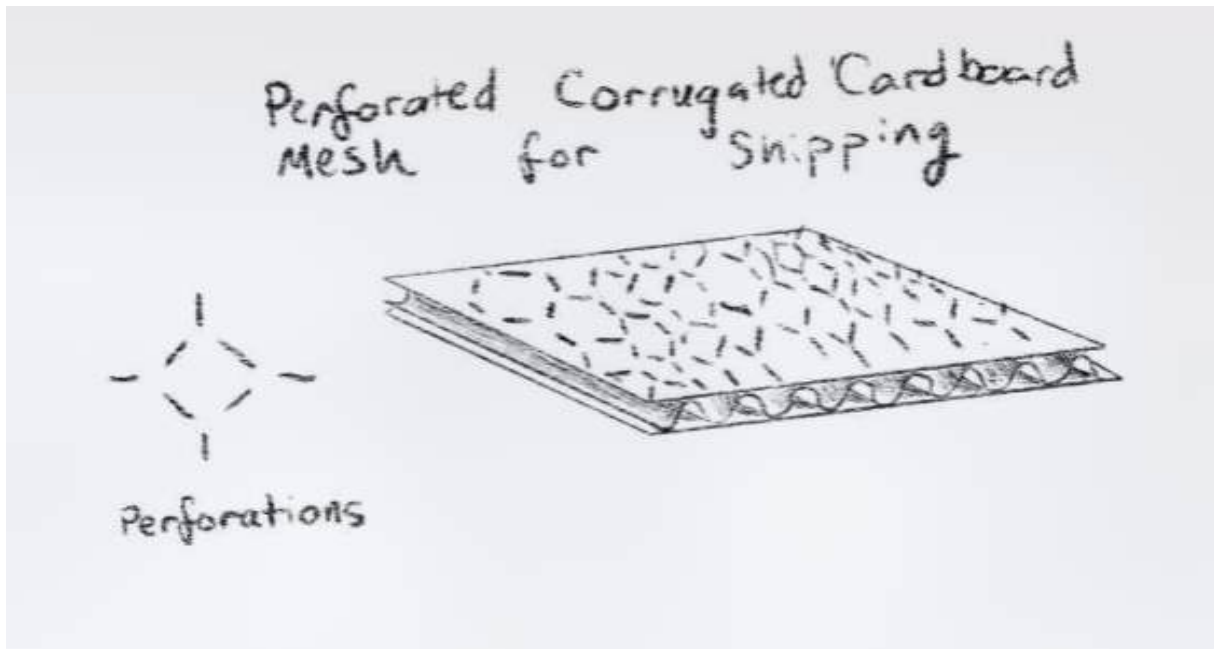


Figure 28: Sketch of Perforated Cardboard Mesh Filler

2.3.4.3 - Looks-Like Prototype



Figure 29: Cardboard Mesh Filler Looks-Like Prototype

2.3.4.4 - Solution to Problem

This cardboard mesh filler provides solutions to a few critical problems that currently exist within the packaging and shipping industries. Currently, there are few options available for protective fillers that are recyclable and reusable. Some existing products in this category include inflated bags and crinkle paper filler. This is a superior product because inflated bags are made with plastic, which is recycled at a rate just 10% of cardboard, and crinkle paper is not convenient to save and reuse at a later time. This product provides a universal (not custom to the user) package-filling solution that is reusable and recyclable.

2.4 - DECISION ANALYSIS

To make the decision about which product was best to proceed with, a few critical factors were considered. These factors were financial viability, feasibility of production, desirability, sustainability, product safety, and versatility. Each product was scored on a scale of 1-5, with higher weights given to more important factors. The two most important factors in the team's decision were the sustainability and product safety provided by the solution. To have the importance of these categories reflected in the decision, these values were doubled in weight on the decision matrix.

Designs	Viability (Financial)	Feasibility	Desirability	Sustainability	Product Safety	Versatility	Score
Weighting	1	1	1	2	2	1	
Suspension Cradle	5	3	4	2	5	4	30
Mesh Insert	5	4	3	5	2	3	29
Textured Cube	4	4	3	5	2	2	27
Dissolvable Mesh	2	3	3	5	4	2	28
Inflatable Cushion	5	4	1	2	4	4	26

Figure 30: Decision matrix

The product that received the highest score was the suspension cradle, which scored well in most categories and was the best performing in the product safety category. In addition to its already high score, there are potential ways to improve the sustainability of the cradle while improving the overall safety of a different solution, which is more difficult to do. For these reasons, the suspension cradle was chosen as the number one contender.

2.5 - PROPOSED DESIGN

The Suspension Cradle is a versatile and innovative solution to the problem that reduces excessive packaging material and ensures product safety. The looks-like prototype of the Suspension Cradle is pictured below (Figure 31).



Figure 31: Suspension Cradle Looks-Like Prototype

The Suspension Cradle consists of an 11.5" x 9" x 5" hinged box, with each half covered by an elastic sheet secured to the sides of the box. The materials used in the prototype are cardboard, tape, synthetic thermoplastic elastomer (TPE), and nails. The materials are subject to change to include a more sustainable design that can be used numerous times.

To use the Suspension Cradle, the box is opened and laid flat. The product or object is put on one side of the box, the box is closed, and the three sides are taped (Figure 32).

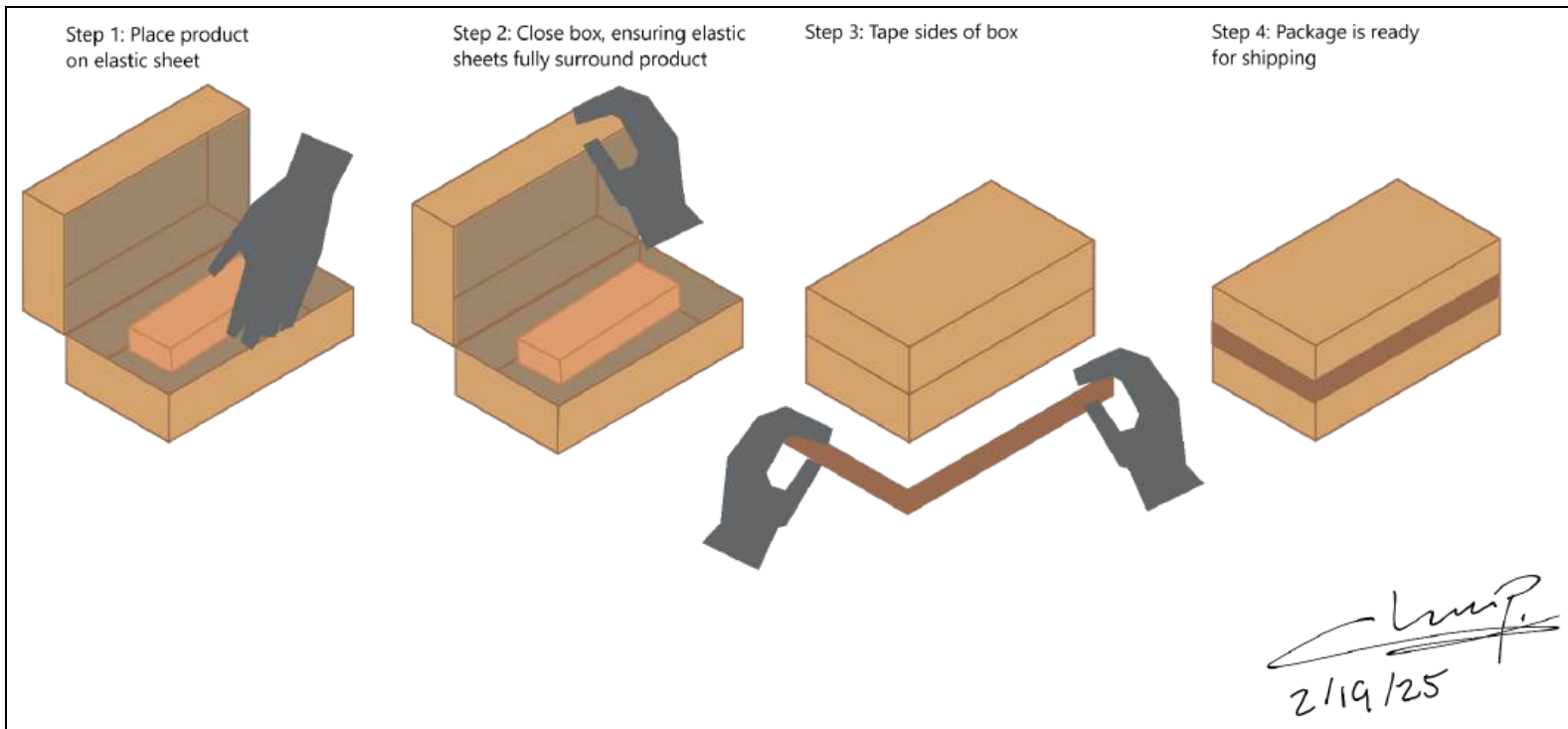


Figure 32: How to Use the Suspension Cradle

Advantages of this product include: improved stability, versatility, load protection, and reduced wear and tear. It satisfies the need for feasibility, product safety, and versatility per the requirements laid out in section 2.2 REQUIREMENTS, CUSTOMER NEEDS, AND TECHNICAL SPECIFICATIONS. Sustainability is the main drawback of this specific design, so the team has started researching sustainable material alternatives to TPE that still provide a large enough friction coefficient to keep the product from slipping inside. Other drawbacks include weight limitations and shape limitations (as products with sharper/pointy edges could puncture the rubber sheets).

2.6 - SUMMARY

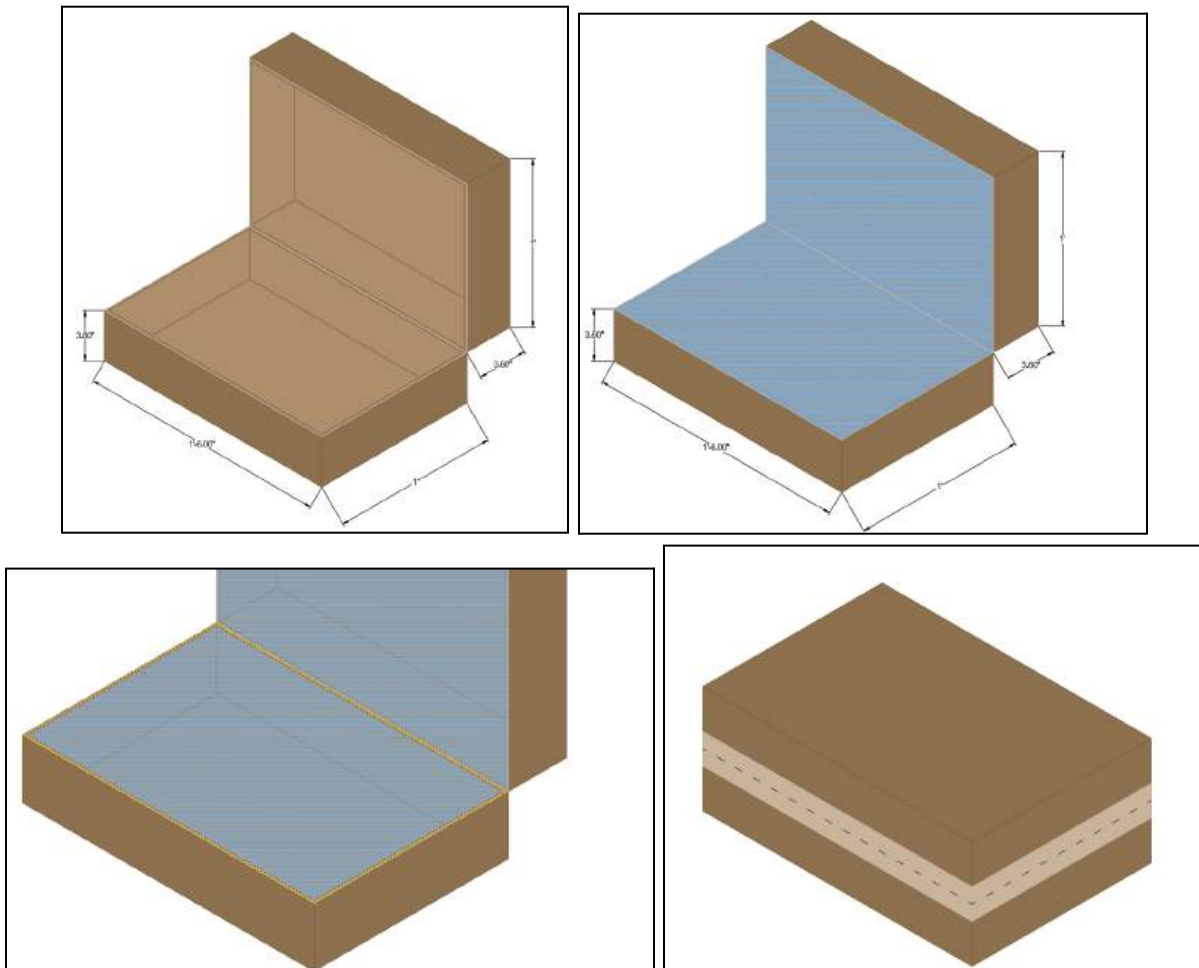
Based on the analysis of the financial viability, feasibility of production, desirability, sustainability, product safety, and versatility of each of the proposed solutions, the Reformers have concluded that the Suspension Cradle is the design that should move forward. The design provides a unique, novel idea that keeps products safe and works towards sustainability in packaging.

3.0 Subsystem Build and Validation

3.1 SUBSYSTEM DESCRIPTIONS

3.1.1 Box Design

The box design subsystem focuses on the visual and functional aspects of the Elastic Suspension Cradle, appealing to the user's interaction and the structural framework that supports the TPE sheet suspension system. This includes determining the overall geometry, dimensions, accessibility, and layout to ensure the proposed solution is appealing and functional. The design prioritizes a minimalistic look with clean aesthetics and symmetry to highlight the “floating” effect of the suspended object, but it must also consider the safety of the object. The final design can be seen below (Figures 33-36).



Figures 33-36: Final design broken down into individual components (box, box with elastic sheet, box with elastic sheet and adhesive, box closed with marked tape)

3.1.2 User Interaction

The purpose of this subsystem is to define and enhance user interaction with the shipping box design. It focuses on three key areas: intuitive understanding of how to open the box, ease of removing the sealant without damage, and clear interaction with the final sealing method. While full user testing is limited without a complete prototype, informed decisions have been made through stakeholder feedback, logical design analysis, and materials research. The final design includes recyclable kraft tape (with water-activated adhesive) marked with dashed lines and scissor icons to guide users, and confirms the "clam design" for ease of use across all age groups.

3.1.3 Specifications and Scalability

The specifications and scalability subsystem define the size ranges for packaging designs. It optimizes packaging for functionality and efficiency while evaluating costs related to size variations. This analysis ensures the packaging solutions are scalable, practical, and cost-effective.

3.1.4 Materials

The purpose of the materials subsystem is to select effective materials that support the chosen design. This includes sturdy cardboard capable of withstanding the tension from the elastic sheets, durable elastic sheets that maintain stretchability, adhesives strong enough to secure the sheets to the box, and tape to seal the package securely. All selected materials must also meet eco-friendly standards, being either recyclable or biodegradable.

3.1.5 Assembly

The assembly and production subsystem encompasses the manufacturing process, assembly sequences, and production methods required to efficiently and consistently produce the protective box and internal cradle system. This subsystem ensures cost-effective and standardized production while maintaining product integrity and meeting quality standards. When scaled up and implemented, this subsystem will include component fabrication, assembly workflow, and quality control measures for commercial manufacturing.

3.2 SUBSYSTEM CONNECTIONS

Each subsystem is connected and must work together to create a cohesive, functional design. The box design creates the structural and aesthetic foundation, which is tied to the user interaction to ensure the product is intuitive, accessible, and looks good. Specifications and scalability align the design with practical size ranges and cost efficiency, integrated in both design and production decisions. The materials subsystem ensures that chosen components, like cardboard, elastic, and adhesives, meet performance and sustainability requirements, and the design must be able to fit within the bounds of materials that are available on the market. The assembly subsystem integrates all the previous decisions into a feasible production process, ensuring the final product can be reliably manufactured at scale while maintaining quality and user experience. Below is a diagram displaying how each of the subsystems is connected (Figure 37).

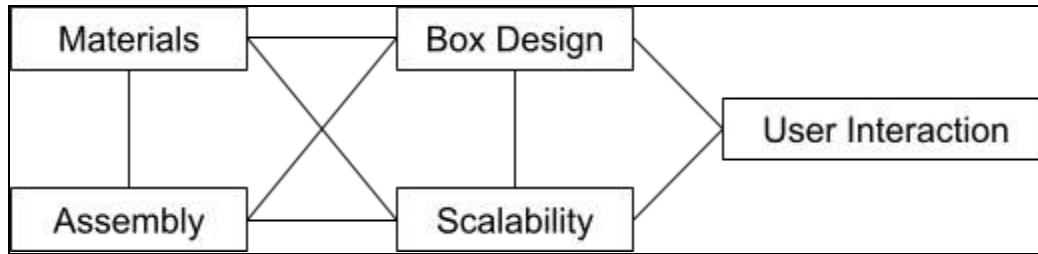


Figure 37: Subsystem interactions

3.3 INDIVIDUAL SUBSYSTEM REPORT SUMMARIES

3.3.1 Box Design

The box design subsystem of the Elastic Suspension Cradle focuses on creating a functional and visually appealing structure that enables an object to be suspended in mid-air using tensioned elastic sheets. The clamshell-style box design was chosen through a decision matrix and stakeholder feedback for its balance of security, ease of assembly, and user interaction. Dimensioning is standardized with the width as a variable input, and height/length are calculated based on that to optimize suspension without damaging the elastic sheets by overstretching them. Solid sheet geometry was selected through a design matrix that analyzed durability, simplicity in fabrication, and ease of integration; despite alternative patterns that would offer more flexibility, the solid sheet design was chosen due to its simple manufacturing process. The subsystem interfaces with others through shared constraints on materials, scaling rules, adhesive placement, and user interaction needs. Final validation came through testing a working prototype, which successfully protected a fragile object across multiple drop tests, confirming the effectiveness of the design.

3.3.2 User Interaction

Through several validation test such as dropping package from 40ft parking garage, showing sketches to users and gaining feedback on user understanding of design, gaining survey data on most effective signage, and doing observational analysis of ink on Kraft tape, it has been proven that the product will remain closed through travel, but when the user receives the package they will not only understand how to open it, but also be prompted through signage (Figure 38) on how to safely retrieve their ordered item. Preconceived user notions of understanding and cultural perspectives, such as a “wedding ring box,” allow the product to communicate to the user a sense of fragility and a need for careful handling when opening. Stakeholder interviews solidified this understanding. Research allowed for a selection of the best possible materials to meet the needs of the subsystem. Eco-friendly ink, along with being water-based, allows for easy recyclability and environmental consciousness of the product, which the user will appreciate. Not to mention the outsourcing of production so that workers in assembly can focus less on the application of ink onto tape and more on overall quality box construction. Tying in with the scalability of the product and overall box design, these pieces of the overall subsystem of user interaction will effectively meet the goal of the product while working in harmony with the other subsystems to meld the world of the designer and user.

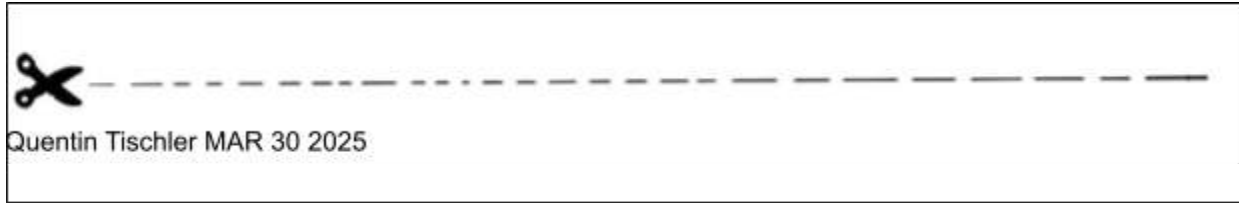


Figure 38: Signage included on the box indicating where to cut

3.3.3 Specifications and Scalability

With the help of other team members, using the physical properties of the chosen materials, the design of the box requires that its height be dependent on the length and width of the box.

Specifically, the height is equal to $\frac{lw}{l+w}$ where l denotes the length and w denotes the width.

Additionally, the cost of this design does not deviate too far from real-world designs and can scale to different sizes. Ultimately, with the work from other team members and feedback from stakeholders, the design of the packaging seems to have real-world potential. The full cost equation can be seen below (Equation 1).

$$\text{Total Cost} = \sum(\text{Adhesive, Cardboard, TPE sheets})$$

$$\text{Total Cost} = \left(\frac{w}{2} + \frac{l}{4} - \frac{1}{8}\right)(0.0102) + 2l\left(w - \frac{1}{8}\right)(0.012) + 4lw(0.00243)$$

$$\text{Equation 1: Total cost equation based on length (l) and width (w)}$$

3.3.4 Materials

The material subsystem focuses on finding the cheapest and most efficient material for the task they are to perform. Through extensive research through secondary sources, the best materials for the box, elastic sheet, inside adhesive, and outside adhesive were recycled cardboard, thermoplastic elastomer sheets, water-based contact cement, and kraft-based water-activated adhesive, respectively. Recycled cardboard was chosen for its cheap cost and relatively strong wooden fibers, the TPE for its rubber-like properties and texture, the water-based contact cement for its powerful bonding properties and spreadability, and the kraft and water-activated tape for its biodegradability and cost. Overall, all product materials were also easy to buy in bulk, allowing a steady source of materials if needed, as well as a reduced price for them.

3.3.5 Assembly

The assembly and production subsystem has been validated through comprehensive testing, stakeholder feedback, and analytical methods to ensure it will meet performance requirements while remaining within budget constraints. Time-motion studies confirmed the efficiency of the assembly sequence, while dimensional accuracy testing verified the consistency of the production methods. Stakeholder feedback from production operators and manufacturing

engineers guided refinements to the workflow and fixture designs. Cost analysis validated that the selected approach can be implemented within the \$100 manufacturing budget, while capacity analysis confirmed the ability to scale production as needed. Together, these validation approaches provide confidence that the assembly and production subsystem will efficiently and consistently produce protective boxes and cradle systems that meet all performance requirements.

3.4 SUBSYSTEM INTERFACES

The Elastic Suspension Cradle design relies on a strong integration between its subsystems: box design, user interaction, specifications and scalability, materials, and assembly. Each subsystem has critical interfaces with others to ensure the final product is functional and efficient to the team's standards.

3.4.1 Box Design Interfaces

With User Interaction: The need for a minimalistic box design sets many parameters toward user interaction, requiring the unboxing to be intuitive while maintaining the visual appeal of a clean, sleek structure. The placement of dashed lines and scissor icons on the tape was designed to complement the overall clean look while guiding the user.

With Specifications and Scalability: The design needed to be able to be scaled to many different dimensions to hold a variety of different objects. This required a need to set a standard equation to calculate the height based on length and width, simplifying the process of calculating costs and material usage.

With Materials: The chosen design requires the need for sturdy, lightweight cardboard for the outer structure, strong elastic TPE sheets for suspension, effective adhesives, and durable, eco-friendly tape. The material requirements were derived from the idea of having a minimalistic, "floating-object" design concept.

With Assembly: The clamshell geometry of the box was specifically developed to allow simple and repeatable assembly. Because the box relies on external sealing rather than internal locking mechanisms, the design had to ensure that after closure, only the final application of tape was needed to fully secure the product.

3.4.2 User Interaction Interfaces

With Box Design: The clamshell box had to not only protect the product but also offer intuitive visual cues for opening. User interaction decisions, like including dashed lines and scissors icons, were made to match a clean, symmetric design and guide users effortlessly.

With Specifications and Scalability: Having clear size options ensured that users could easily select a packaging size that fit their product without excess material or confusion. Consistent interaction design across all sizes reinforces ease of use.

With Materials: Materials played a smaller role in the user's experience; however, each of the materials chosen contributed to a perception of both durability and eco-responsibility, enhancing user satisfaction.

3.4.3 Specifications and Scalability Interfaces

With Design: Scalability interfaces helped set the geometric constraints for the box design subsystem, ensuring that the clamshell design could be proportionally scaled while maintaining functionality and aesthetics. This allowed the design to adapt to different product sizes without losing structural integrity or user appeal.

With User Interaction: The availability of multiple box sizes through scalability ensures that end users can select appropriately sized packaging for their needs. This ensures user satisfaction by offering practical options while maintaining the cheapest design for their needs.

With Materials: Specifications and scalability determine the appropriate material dimensions and range needed to accommodate various package sizes. This data helped identify materials that could meet strength, stretch, and durability requirements and reasonable costs. Cost estimates for recyclable Kraft tape, cardboard, and TPE sheets were based directly on size specifications, ensuring scalability.

With Assembly: The defined box dimensions formula informed the assembly subsystem by providing consistent sizing standards. This enabled planning for streamlined assembly processes, minimal tooling requirements, and easy scalability to mass production.

3.4.4 Materials Interfaces

With Box Design: Material properties needed to be lightweight, stiff, stretchable, and have enough surface friction to hold the object inside. The materials had to support the structural and aesthetic goals, particularly ensuring the box was lightweight yet strong enough to protect the suspended object.

With User Interaction: Eco-friendly materials (recyclable or biodegradable) were chosen to enhance the user's perception of sustainability. Clear physical feedback from the materials (such as firm but easy-to-cut tape) also improved the usability of the package opening process.

With Specifications and Scalability: The materials needed to be able to scale with a low cost, and needed to be able to adjust to the scaling dimensions. This required finding materials that were sturdy, stretchy, and durable.

With Assembly: The cardboard had to be either prefabricated into clamshell cut forms or formed on site using specialized equipment, TPE sheets required precision manufacturing (ideally 3D printable to meet elasticity and size specifications), and adhesives needed to be easily applicable during assembly and minimize waste. Material characteristics, such as bonding strength and durability, determined how each assembly was structured.

3.4.5 Assembly Interfaces

With Box Design: Assembly sequencing was influenced by the clamshell box design, requiring jigs that could handle this type of box. Box design needed to be simple to allow for continuous flow production, minimizing tool use and error rates. The elastic mounting had to be able to be done by machines with precise, tight fits.

With Specifications and Scalability: The assembly process needed to accommodate different-sized boxes effectively. Standardizing some steps while allowing flexibility for different sizes was critical for maintaining cost effectiveness and quality across all variants.

With Materials: Chosen materials needed to be able to be applied efficiently with precision and control for consistency across all variations of box sizes.

3.4.6 Whole System Interaction

- Box Design provides the physical structure and user-facing aesthetics.
- User Interaction ensures the product is easy, intuitive, and satisfying to use
- Specifications and Scalability ensure the design can accommodate a range of products cost-effectively.
- Materials are selected to match structural, environmental, and usability goals.
- Assembly turns the design into a practical, manufacturable product with minimal waste.

4.0 Team Value Proposition and Risk Report

4.0.1 Reference Sizes

Because the design is scalable, stakeholder feedback and research were used to identify three standard box sizes capable of accommodating a wide range of objects. The standard box sizes referenced throughout include:

Reference Name	S1	S2	S3
Size	10"x10"x5"	14"x14"x7"	12"x18"x7.2"

Table 1: Box Sizes Referenced in Module Report

4.1 SOLUTION VALUE PROPOSITION

4.1.1 User Interaction

The sustainable packaging solution helps to eliminate the problem of environmental damage by replacing traditional packaging solutions with an improved clamshell-style box that has an enhanced potential for recycling and reuse while meeting or exceeding performance standards of traditional packaging. (Stryder)

After looking at the possible risks, the benefits outweigh the costs when considering the following:

- Environmental Impact Scale: When compared to traditional “safe” packaging solutions like styrofoam and foam, this product is much more sustainable. Styrofoam can take 500 years to degrade and has limited potential for reuse, while this product would both degrade much faster and have more opportunity for recycling and reuse.
- Value from Market Differentiation: The product is novel and unique and has the potential to be made in both “economy” and “luxury” versions. This gives a market advantage to the clamshell box design because it can satisfy a variety of customers. Along with the trend of sustainable designs, prices for eco-conscious designs won't shock the consumer.
- Box System Design Optimization: For efficient production, a new assembly line has been designed so that it could potentially integrate into the manufacturing process of other products.
- Cost when compared to traditional packaging solutions: The product is likely to cost very similarly to traditional packaging solutions, and because of the other value-adds of the product, the clamshell box design is more desirable.

(Stydr (edit and minor addition: Quentin T.)

The key parts or primary concern of the user will be opening the package and following the instructions on the package. HP Latex Ink, while more costly than others, sports a recyclable chemistry along with a guarantee of clean quality printing needed to survive shipping. Meaning the instructions on the tape and box will survive transport. A quality recyclable Kraft tape, devoid of excess plastics or non-recyclable fibers, is also needed as it is the final sealant that

keeps the whole box shut. With these needs in mind, it is evident that these two materials, despite being more specialized and greater in price than their common cheap counterpart, rationalize a value exceeding the increase in pricing. (Quentin T.)

4.1.2 Materials and Cost for Works-like Prototype

(All)

Below is a table referencing the materials used in the works like prototype and the costs associated with each piece (Table 2):

Item (S2)	Unit of measure	Cost per Unit	Quantity per Prototype	Total Cost
Single-Walled Corrugated Cardboard	S2 Sheet/Box (14"x14"x7")	\$2.09	1	\$2.09
69% Polyester Fabric Sheet	Square Inch	\$0.042	192	\$8.00
Recyclable Tape	Foot	\$0.07	3.5	\$0.25
Fabric Glue	Fluid Ounce	\$4.43	≈ 0.6	≈ \$2.66
Sharpie	Square Inch	≈ \$0.0014	≈ 1.0025	≈ \$0.0014
Total Cost for Works-Like Prototype				≈ \$13.00

Table 2: Materials and Costs for Works-Like Prototype

4.1.3 Materials and Cost for Full-scale Final Product

(All)

Below is a table referencing the materials that will be used for the final product and the predicted costs associated with each piece. The medium-sized box is the cost reference for all following data (14"x14"x7") (Table 3):

Item (S2)	Unit of measure	Cost per Unit	Quantity per Product Unit	Total Cost
Single-Walled Corrugated Cardboard [15]	Square Inches	\$0.9	1	\$0.90
TPE Rubber Sheet [16]	Sheet	\$0.47	2	\$0.94
Water Activated Kraft Tape U-line [17]	Foot	\$0.01	3.5	\$0.04
Water-Based Contact Cement [18]	Milliliters	\$0.03	3	0.09
HP Latex ink Black [19]	1.23 mL/sq foot ink	\$0.12	.0885(sq ft)	\$0.01
Total Projected Cost for Realized Product				\$2.19
Manufacturing and Upcharge Costs		20%		0.44
Total Projected Price for Realized Product				2.63

Table 3: Materials and Costs for Realized Product (Jack Morris)

Based on this table, the total cost for a size S2 box would be \$2.19, and the total price would be \$2.63. Using the same math for the S1 and S3 boxes, the total projected costs come out to \$1.26 and \$2.35, respectively. Next, with the manufacturing and upcharge costs, the total price of the S1 and S3 sizes comes out to \$1.51 and \$2.82, respectively. (Jack Morris)

4.1.4 Benefits

This design provides a high level of safety for moderate to small-sized fragile glass items without sacrificing cost or user friendliness. With current packaging options, glass is typically wrapped or submerged in mass-produced packaging materials, such as packing peanuts, bubble wrap, multiple boxes (russian doll), or plastic air pillows [20] and then shipped in regular boxes that are ill-suited for the task. In the rare case that glass products are given a specialized box or packaging, the cost is greatly increased and oftentimes has substantial packaging waste once the delivery has been completed.

The primary idea behind this design focuses on trying to improve user interactions without compromising safety or price. With this idea in mind, the inside elastomers in this design were specially chosen to be able to hold a wide variety of products, focusing on a one-size-fits-all style package, removing the need for specialized boxes in the first place and allowing it a wide applicability scale. This design also can be used multiple times, allowing the user to repurpose the box for numerous personal shipping needs, improving monetary value to the consumer. Also, the box itself has clearly outlined instructions, allowing anyone to be able to understand how to properly use the box without having to understand the entire design. Once the user is finally done with the box, or when the design outlives its life, it can be easily recycled and reused for future boxes, as all parts of the design can be recycled and/or repurposed. (John D.)

Having greater versatility plays into a downgrade in material costs by eliminating the need for excessive protective fillers like packing peanuts or molded foam.

Stemming into the boxes inherit “design for mass production,” which further reduces cost. By automating the process through human workstations or automated machines, there are several benefits, such as reduced chance for error, greater uniformity, and cost reduction.

The subsequent removal of single-use plastic waste and replacement with recyclable materials in the design enhances consumer perception of the product. This allows consumers to see themselves as getting something they wanted while simultaneously helping the Earth.

Even with all these cost mitigation strategies in place, the realized product will cost more than a standard box packed with the associated single-use waste products. Despite the higher cost, though, the reduction in the number of package failures that end up damaging a product and the subsequent compensation a company would have to afford are offset by the design.

Overall, this product’s minimalistic, durable, and recyclable design provides long-term economic and environmental value that surpasses traditional packaging options. (Chase)

(The above pieces were edited and had minor additions by Quentin T.)

4.2 RISKS AND MITIGATIONS

(All)

Below is a table of all the risks associated with the Suspension Cradle solution (Table 4). Each risk is categorized with the likelihood of it happening, the impact it will have on the user or product, and the overall magnitude of the risk. Each risk has a proposed mitigation strategy to offset the chance of it happening.

Risk	Likelihood	Impact	Magnitude	Mitigation
TPE Sheet Ingestion	Unlikely	Moderate	LOW	A label will be applied to the outside saying “Keep away from children.”
TPE Sheet Tear	Unlikely	Moderate	LOW	Tensile strength and Elmendorf tear tests will take place with each batch of TPE sheets
TPE Sheet Over-tensioned	Unlikely	Major	MEDIUM	Tensile strength and Elmendorf tear tests will dictate the fragility of items that can be placed inside
Incorrectly Applied Tape	Unlikely	Major	MEDIUM	Labels will be added for tape placement
Inability to be Properly Marked and Labeled	Unlikely	Major	MEDIUM	The cardboard surface must be unobstructed, and labels must be able to adhere to the outside
Opening Package Incorrectly	Likely	Major	HIGH	Labels will be added for opening instructions (placement, where to cut, how to open clamshell)

Table 4: Possible risks categorized with likelihood, impact, and magnitude of risk, along with mitigation measures (Chase/Jack)

Several risks present themselves in the realized solution, which have been identified in the risk matrix above. The predominant number of risks fall into the lower and medium magnitudes but still need to be addressed. For users, the most significant risk is improper opening of the box, which could cause the suspended product to slide out and fall. This is classified as a high-risk issue due to its likelihood and impact. To mitigate this, the team proposes applying clear warning labels and instructions on the exterior of the box to guide users on how to safely open the package. One other low-risk issue is the ingestion of the TPE sheet by children or animals. The TPE sheet is hard to tear and is not small enough to be swallowed; however, there is still a risk associated with this. To mitigate this risk, the solution will be marked with a “Keep away from

children” label. From the manufacturer's point of view or a consumer using the Suspension Cradle to ship a personal item, there are a few more identified risks. The risk of incorrectly applied tape during sealing or resealing could present the possibility of the box opening and the item inside falling out. This is a medium risk and will be mitigated by adding labels for tape placement. The risk of improper shipping labeling presents the possibility that a package gets lost in transit. To mitigate this, there must be a clear spot as to where a shipping label should be placed; this spot must be unobstructed and on a side of the box where it can be most easily read. As for TPE sheet risks, there is the possibility of the sheet being too tight and damaging the product inside (medium risk) and the possibility of the TPE sheet ripping (low risk). Both of these risks require proper testing from the manufacturer to make sure the tensile strength is not compromised, too tight, or too loose. To test the properties of the sheet, Elmendorf and tensile strength tests will be conducted before the sheet's installation into the box. (Chase/Jack)

(Module 4 formatting and revisions: Chase/Jack)

Citations:

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5.0 Proposed Full-Scale Solution

5.1 FINAL CONCEPT DESCRIPTION

The Elastic Suspension Cradle (ESC) is a “clam shell” style box of corrugated cardboard with two TPE elastic sheets inside that, when closed, suspends a fragile piece of glassware or comparable item, keeping it from shifting during shipping and protected given a fall or drop. This design has proven to be successful in product protection up to a minimum height of 40ft. The 100% recyclable design means that even after its use has passed, it is still a net positive through the end of its product life cycle.

5.1.1 Cad Model and Technical Drawings

Below is the final design model in an isometric view, it does not have dimensions attached as it is scalable to various dimensions (Figure 39)

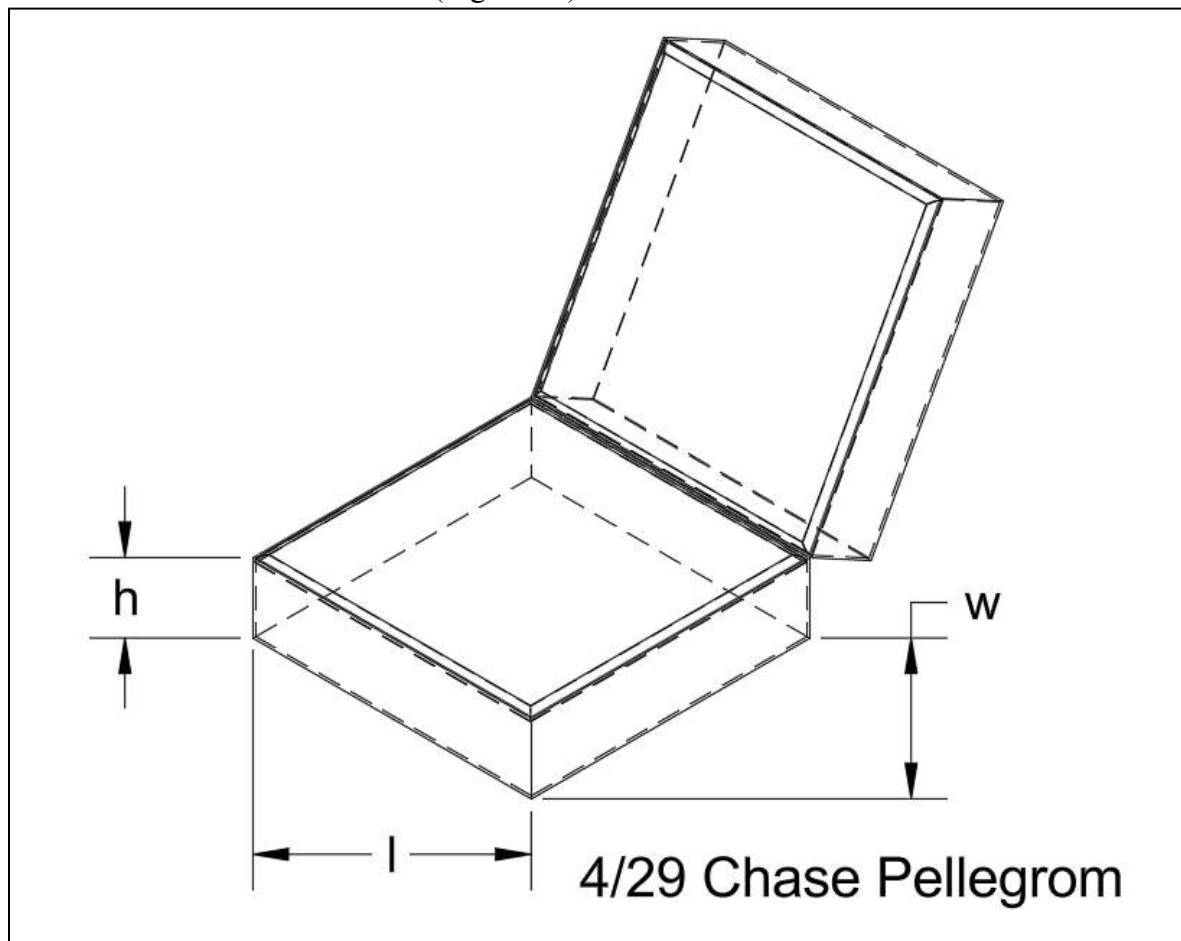


Figure 39: The final model includes the box and TPE sheets with (h) being height, (l) being length, and (w) being width.

5.2 FUNCTIONALITY, COMPONENTS, AND PHYSICAL PROPERTIES

5.2.1 Proposed Materials and Their Properties

Within this design there are four different materials needed to create the final product. These four materials are the cardboard, the adhesive, the inside liner sheets, and the tape. Starting with the cardboard, this design will utilize recycled cardboard due to its cheap nature, moderate strength, and eco friendly name. The adhesive will be a water based contact cement. The reasons for choice are that water based contact cement can easily be recycled and is biodegradable, can spread easily allowing maximum usage of the cement, and can easily grab onto all manners of material which is necessary when dealing with elastic materials. For the inside liner sheets, this design opted with a special thermoplastic elastomer named FX-1515 that's sold by Green Dot Bioplastics. The material of TPE was selected due to its 3-D printable nature and its elasticity. The specific type of FX-1515 was chosen due to its high friction component, its low cost, and its ability to stretch to almost 700% of its original form. For the specific tape used, a craft base with water activated adhesive was chosen as the best option. Both the craft base and the water activated adhesive would be perfect for recycling, and would allow a strong hold on the box to keep it shut while shipping. A prime example of this tape would be U-Lines' water activated packing tape.

5.2.2 User Interfaces

The user interfaces of the Elastic Suspension Cradle include the opening mechanism of the box, the removal of the Kraft tape sealant, and the retrieval of the suspended product. Visual indicators, such as dashed lines and scissor icons printed on the tape, serve as guidance tools to make the opening process clear and intuitive. Additionally, the overall shape, size, and material feel of the box contribute to the user experience, ensuring that interaction with the package is simple, efficient, and user-friendly.

5.3 CONCEPT VALIDATION

5.3.1 Authoritative Research

Several components had to be researched before finalizing a design. The first component was the geometry of the elastic suspension and the fastening mechanism. Based on research in flexible membrane structures and kirigami mechanics, the geometry of the elastic suspension sheet plays an important role in determining its mechanical purpose. Solid sheets, while structurally strong and simple to fabricate, offer limited stretchability and can develop stress concentrations when deformed. Sheets with cut patterns (kirigami designs) can significantly increase both elongation and flexibility by redistributing strain through controlled deformation points. Studies have shown that kirigami-cut thermoplastic polyurethane (TPU) sheets can achieve up to 218% elongation while also reducing stiffness, making them highly adaptable for suspension and shock-absorbing applications [1]. More complex hybrid kirigami patterns have demonstrated even greater extensibility, achieving up to 750% strain and reducing stiffness by a factor of 30 compared to uniform patterns [2]. Lattice or mesh-based geometries are the most flexible and efficient in

energy absorption, though they typically have low raw-bearing strength. These open geometries are extremely effective in distributing tension and minimizing peak stresses, making them ideal for packaging fragile or lightweight objects [3], [4]. Despite the obvious advantages of these cut patterns, they complicate the ease of fabrication quite a bit. Not only do you have to cut out the shapes, but you have to adhere the many separated pieces to the sides of the box in particular ways. This makes it difficult to advertise as an easy and convenient alternative to typical packaging materials (bubble wrap, paper, etc.).

The second key component was the mesh itself; thermoplastic elastomer (TPE) was the best of all of the potential materials that could have been used. This was due to its low cost, high flexibility, and great friction coefficient. Overall, it was stated that TPE could stretch to almost twice its length and still be undamaged, which would be a critical factor in preventing products from being too squished or forcing a custom insert to be made every time a new product was shipped inside. Additionally, TPE is known to be a soft material with a grippy texture (being referred to as printable rubber), meaning that once the box is closed, the products inside will not move around or scratch [5][6]. While TPE can be confusing since it's both an overarching category as well as a product, the product itself was the best fit. The other options of thermoplastic polyurethane (TPU), copolyester (TPC), and polyamide (TPA) are also good in terms of durability but lack the elasticity and texture that's needed to fully protect the products within. Examples of this are that TPU grants a higher temperature resistance and a stronger ability to withstand deformations after stretching, but has a rougher texture that could potentially scratch glass during transport [7]. TPC is the heavy-duty application of TPE filaments, boasting an extremely high durability, sterilization compatibility, and resistance to most chemicals or oils. However, because of all of this, TPC has a very low elasticity compared to the original filament, as well as has a much rougher texture [8]. TPA is another heavy-duty filament, but it has less resistance than TPC. Instead, TPA boasts a high heat resistance, a high durability, and a moderate elasticity when compared to the original TPE. The only incompatible aspect that clashes with the design is that TPA tends to warp fairly easily, meaning it's not suited for an environment where it will interact with a variety of factors that could arise during shipping [6].

For the cardboard used in the boxes themselves, it was decided to use recycled cardboard rather than virgin cardboard to decrease costs and improve eco-compatibility. The difference between virgin and recycled cardboard is the weakened wood fibers in recycled cardboard due to the extensive pulping and chopping processes it has to go through to be reused. Once a cardboard sheet is recycled, it is sorted into cardboard piles, shredded, and mixed with water to create a pulp, filtered to get out the impurities and ink, and laid down in new sheets to dry into new products [9]. While this may mean the recycled cardboard is weakened, the positive environmental impacts, such as reduced water usage, reduced resource usage, and decreased waste output, make the recycled cardboard a much better alternative. Not only will this help with costs, but it will also help with public opinion of the product because they will feel like they are helping the environment by using the product rather than hurting it [10].

For the adhesive, water-based contact cement is the best option based on cost and strength of the adhesive itself. Compared to other glues, contact cement is a great choice due to its strength, versatility, and ability to grab onto a variety of different materials. All of these are important when considering not only what the cement is going to be used for, but also the load it is

expected to hold. The choice between solvent and water-based contact cement was based on the water-based contact cement's viscosity, increased strength, and “green” name based on its lack of solvent. The viscosity of this glue is important because it will need to cover a long stretch of TPE, and if too much is added in one place, it can be spread out to increase ease of manufacturing. The strength is important because the loads that the design might carry could be a range of weights, so all weights must be possible to prepare for the worst-case scenario [11].

For the outside adhesive, the best possible adhesive would be an eco-friendly tape with a kraft paper base and a water-activated adhesive. By including a Kraft paper base, the tape is not only sturdy and tear resistant, but also biodegradable when it's done being used. As for the particular adhesive type, water-activated adhesive is the best possible option for simplicity (dab of moisture to activate gum adhesive) and effectiveness. There is a possibility of a water based adhesive would possibly the tape would prove to be too much of a hassle to put on and could potentially cause tearing in the box once bonded and then removed, preventing reusability of the box [12]; however, this is unlikely as it is still being sold by the main line distributors such as Uline which do not mention avoiding use on cardboard.

5.3.2 Stakeholder Feedback and Interaction

The design process involved multiple outreach efforts, with two major ones validating the final product. The team gathered feedback on sustainable design from various individuals, focusing on what they appreciated or disliked about modern trends like paper straws and compostable silverware. A common concern was that sustainable products often compromised quality and effectiveness. Stakeholders indicated they would accept higher costs for sustainable options, but emphasized that functionality is more important. Along with that, the concept of a clamshell box received positive feedback when presented as a sketch. This validated the core design, allowing the team to move forward.

Stakeholder input also influenced the tape design. A survey across various age groups identified the scissors and dashed line symbol for indicating the cutting line, which was incorporated into the final design. This symbol guides users on how to open the box without damaging the critical back fold, which protects the contents. The feedback from the stakeholder interviews gave us clearance on the design and allowed us to move forward with testing.

5.3.3 Physical Testing

To ensure that this product was capable of protecting objects inside just as well as current market solutions, the team conducted several drop tests from the McNeil Parking garage in Golden, Colorado, approximately 40 feet high. The typical storage shelf for a shipping depot ranges from 8 to 12 feet high; however, some can go up to 20 feet for storing pallets [13]. Dropping the design from 40 feet allowed us to conclude that it could withstand a drop from over twice the maximum height this box will ever be stored at. After the first two drops, the team noticed that one key component began to fail: the clamshell box hinge. Since it was not supported with tape, the box began to tear, starting from the edge. To fix this, the team implemented backfolds on the hinge during the assembly process that would prevent the break starting from the edge.

5.3.4 Analysis and Calculations

With the different sizes the end product of this box is supposed to have, this product has a continuum of different sizes that the box can be. Due to the elastic properties of the TPE sheets, research has shown that these types of elastomers can stretch between 200% to 700% before the sheet tears depending on the conditions and environment the sheet is in [1]. Since the sheet tearing is bad for the protection of the product that is being shipped the team decided to assume the minimum elasticity of the TPE to ensure the protection of the product. Using this information the height of the box can be determined by the length and width where $h = \frac{lw}{l+w}$. Using this, the cost of each material per square inch and the geometry of the box itself, the total cost equation comes out to:

$$\text{Total Cost} = \frac{w(0.0102)}{2} + \frac{l(0.0102+0.00243)}{4} + 2lw(0.00243 + 0.024) - \frac{(0.0102)}{8}$$

where l and w represent the length and width of the box respectively. After calculating the total cost of the product a total of a 40% upcharge is added to the price so that it covers manufacturing costs and profit for the team. One example of this equation in action is the works-like prototype had dimensions of 10x10x5” so the total cost would come out to \$1.51 and the total price would be \$2.11. So by using the equation listed above, if someone needs to produce a different size for this design they can use the equation to calculate the total cost and price of this box.

5.4 Conclusion

The initial problem statement that the Reformers attempted to solve was: “How might we reduce the waste produced by single-use packaging materials?” The Elastic Suspension Cradle (ESC) represents a large advancement over traditional packaging solutions. By incorporating sustainable materials with a novel and innovative suspension mechanism, the ESC both protects fragile items through multiple impacts and is completely recyclable at its end-of-life. The design succeeds in three critical ways: manufacturability, allowing for cost effective production at scale; sustainability, minimizing environmental impact throughout its life; and affordability, ensuring viability in a competitive market. As demonstrated through testing and analysis, the ESC effectively addresses the waste reduction challenge while maintaining complete product safety. This positions the ESC as a practical and environmentally ethical alternative to traditional single-use packaging solutions.

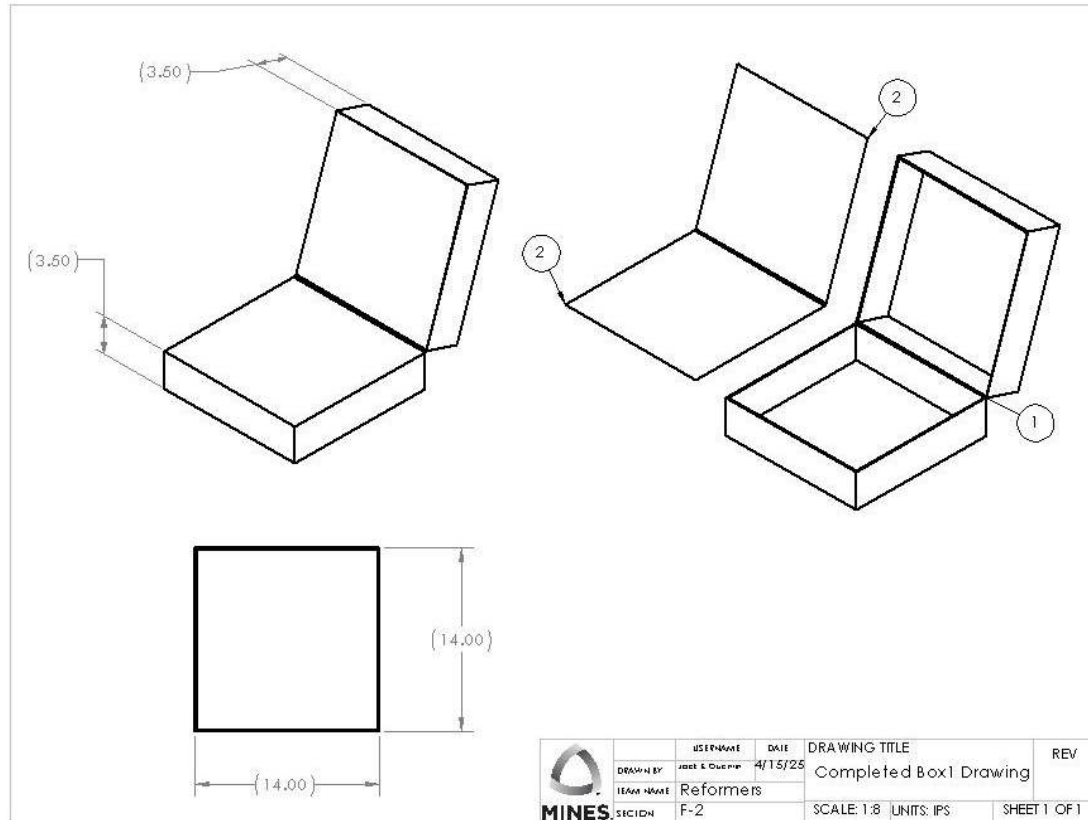
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5.6 Appendices

5.6.1 Working Drawings



5.6.2 Team profile

Name: Jack Morris

(Intended) Major: Computational Applied Mathematics

Expected Graduation Year: 2028

Experience/Background: Know how to use Python, LaTeX, and currently learning C++, and finished various math courses such as calculus III, differential equations, and linear algebra.

Aspirational Design Goals: Figure out a more sustainable way to use packaging that minimizes overall waste.



Name: Strydr Silverberg

(Intended) Major: Computer Science, CENG

Expected Graduation Year: 2026/2027

Experience/Background: Experience with programming, specifically Zig, C, C++, Docker, Kubernetes, Python, and LaTeX. I enjoy chess programming and I'm a proponent of Open Source software.

Aspirational Design Goals: I want to create a device that allows users to play against my chess engines in real life.



Name: Quentin Tischler

(Intended) Major: Environmental Engineering

Expected Graduation Year: 2028

Experience/Background: 5 years of design experience and technical work in theater, including other subjects.

Aspirational Design Goals: "Some of my design goals are increasing sustainable design while reducing its impact on the consumer, having an aesthetically pleasing solution, and a well-put-together report."



Name: John Davis

(Intended) Major: Biochemistry

Expected Graduation Year: 2024

Experience/Background: Multiple years in team sports and activities, extensive understanding of tools/machinery for woodworking

Aspirational Design Goals: To learn the process of creation, refinement, and patents for future projects that I hope to work on.



Name: Chase Pellegrom

(Intended) Major: Materials Science

Expected Graduation Year: 2028

Experience/Background: High school architecture classes/competitions, Solidworks design, coding, AutoCAD

Aspirational Design Goals: Create a probable solution to a genuine problem

